Full proof supplement to The exact maximum growth factor for complete pivoting on real 5×5 matrices

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Abstract

This supplement supplies the detailed reductions, source inequalities, finite exclusion specifications, and transport data accompanying the main manuscript. It retains the full technical appendices of the expanded proof and the complete cross-saturated source cover, while avoiding a second copy of the main exposition. The mathematical claims and the stated scope of the exact computations and partial Lean formalization are unchanged.

How to read the two documents.. The main manuscript contains the theorem, the complete elimination coordinates, the principal analytic arguments, and the rules that turn the finite certificates into a global bound. This supplement is part of the material to be reviewed with it. Appendices A–F have their own equation and theorem numbering. A reference prefixed "M" points to the main manuscript; a reference prefixed "FP" in the main manuscript points here. Notation and the normalization are those of the main manuscript. The separate Supplementary Index S1–S8 locates the executable certificate payloads, dependencies, and verification records. The original expanded manuscript is retained in the editorial archive; this rearrangement introduces no new mathematical claim or verification run.

Appendix A. Proof modules and supplementary material

The Supplementary Index, parts S1–S8, maps the analytic and computational dependencies by their mathematical role. The theorem statements in the text specify the domains on which each component is used.

Supplementary unit	Mathematical content and role
S1: Constant and attainment	The degree-61 polynomial, exact isolating interval, identity with the constant of [2], and supplementary algebraic reconstruction of an attaining matrix. The lower bound is [2, Theorem 2.2].
S2: Scalar bounds and complete coordinates	The conditional three-pivot envelope, the first-four-pivot bound, the complete two-step lift, and the same-height D/X saturation proved in Sections M3–M5. The saturation statement used here requires positive fifth-pivot height.
S3: Canonical representative and height	Qualified representative existence, exact row-scale reconstruction, and canonical localization and height certificates, established independently of the final negative-diagonal bound. Six complete exclusion trees contain 228542 nodes and 114274 contradiction leaves. The subsequent 152467-record localization uses 208274 contraction certificates; its three terminal boxes are covered by the algebraic height theorem.

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Supplementary unit	Mathematical content and role
S4: Complete cross-saturated domain	The four source domains and exact sign and transpose maps of Proposition Appendix E.1, including their shared boundaries. The low-r component uses S3. All statements concern complete real matrices; no rank, cofactor-sign, or auxiliary entry assumption is added.
S5: Negative-diagonal framework	The three complete feasible paths, attained six-variable maximum, and seventeen-variable root of Section M7. Zero factors, capacity ties, and the proper negative-diagonal range $k \leq 2$ are retained.
S6: Exact acceptance rules	Rational interval and nonnegative-combination certificates, qualified contractions, algebraic-height rules, and complete feasible transports. Contradictions above γ and safety at α have the distinct semantics of Lemma M8.2.
S7: Complete negative-diagonal cover	The accepted base proof, complete parent proofs, and exact source bindings of Section M9. The final 54-terminal source completes the last parent's 240/240 responsibilities and the total 469/469 original parents. Four source-bound completions cover the base tree's four retained open records; no original source region remains uncovered.
S8: Reproduction and dependency index	Model and payload identities, acceptance records, executable dependencies, and source-composition maps. The final composition's 357 direct dependencies are supplemented by their recursive dependencies; the direct list alone is not the complete proof archive.

The dependency direction is S3 to S4, and then S4 to the cross-saturated safety rules used in S6–S7; the final negative-diagonal cover is not an input to S3 or S4. Counts identify accepted objects, while their rules and exact source bindings establish validity and coverage. An α-safe region may contain feasible matrices.

Component acceptance and final composition are separate operations. The latter reuses identified acceptance records and checks their source bindings; it does not rerun every component's mathematics. Accordingly, the completed composition is not claimed to be a new from-scratch replay of the entire dependency chain. The supplementary index distinguishes the recorded acceptance runs, the checks performed during composition, and the materials required for independent reproduction.

Appendix B. Canonical representatives and the exact near-peak height

Appendix B.1. The original source and the precise reduction inputs

This appendix separates three operations: choosing a representative of a prescribed attainable height; reconstructing that representative from one set of row-scale coordinates; and bounding the resulting height by a constrained maximum argument. All three operations are developed below. The common-column budgets, wall recovery, and prescribed-target construction have explicit analytic proofs; the safe-tail, stationary-source, exceptional-contact, and localization exclusions are identified separately as finite computational inputs. This ordering prevents the final X or negative-diagonal bound from being used to justify an earlier near-peak terminal.

Reading map.. The deduction proceeds in the order below. Each representative or reconstruction retains a common physical source and its stated qualifications. The last column distinguishes analytic arguments from finite certificate inputs; their source inclusions are proved before acceptance is used.

Sections	Conclusion used next	Method and finite inputs
Appendix B.2	High-value budgets and simultaneous first/middle columns	Analytic recovery; thirteen fixed safe-tail positivity checks.
Appendix B.3–Appendix B.5	Same-head wall maximum and qualified head entrance	Tail reconstruction and maximum argument; C19 and difference exclusions.
Appendix B.6–Appendix B.7	Representative at a prescribed height; endpoint equivalence	Exceptional-contact exclusions, compactness and minimum attainment.
Appendix B.8	One complete row-scale coordinate system	Analytic reconstruction in both directions.
Appendix B.9–Appendix B.10	Localization and a compact model near the sharp height	Finite localization, qualified target transport and upper-target exclusion.
Appendix B.11–Appendix B.12	Forced contacts at a maximum and the exact height α	Uniform Jacobian bound, feasible local variation and algebraic elimination.

Section Appendix B.13 is the source and certificate crosswalk, not a further deduction. The chain runs from canonical source to exact height; neither the later complete X bound nor the final negative-diagonal theorem is an input to it.

All indices of the 3×3 blocks and their vectors are $0, 1, 2$. The letter s in this appendix means $p - 1$, and R means the scalar $F/2 - 1$. They are not the tail coordinate s of the negative-diagonal fibre or the real field $\mathbb{R}$. We write $\Gamma = V - \beta Y$ for a cofactor, reserving the threshold γ of the main paper for $4132517/10^6$.

Definition Appendix B.1 (Complete canonical source)**.** Take real parameters

$$(t, E, \beta, V, T, X, Y, Z, p, k, r, w, a, b, g, h, v, j, q)$$

and set

$$\begin{gathered} u = (1, V, T)^T, \qquad x = (X, Y, Z)^T, \\ \widehat{v} = (-g, h, v)^T, \qquad \widehat{q} = (-j, -1 - \beta h, q)^T, \\ D = \begin{pmatrix} k & -a & -b \\ tk & r - ta & r - tb \\ -Ek & r + Ea & w + Eb \end{pmatrix}, \qquad S = D + x\widehat{q}^T, \quad O = S + u\widehat{v}^T, \\ P = \beta\widehat{v} + \widehat{q}, \qquad L = px - u, \\ M = \begin{pmatrix} 1 & -1 & \widehat{v}^T \\ \beta & p - \beta & P^T \\ u & L & O \end{pmatrix}, \qquad F = r - w. \end{gathered} \tag{B.1}$$

The source belongs to $\mathcal{K}$ when

$$\begin{gathered} p, k, r > 0, \quad |t| \le 1, \quad 0 \le E \le 1, \quad 0 < \beta, T, X, Y, Z \le 1, \quad 0 < V < 1, \\ 0 \le a, b \le k, \quad |w| \le r, \\ |p - \beta| \le 1, \quad |L_i| \le 1, \quad |\widehat{v}_j| \le 1, \quad |P_j| \le 1, \quad |O_{ij}| \le 1, \\ |\widehat{q}_j| \le p, \quad |S_{ij}| \le p, \quad |D_{ij}| \le k, \end{gathered} \tag{B.2}$$

for every indicated index, and all six cofactors

$$\begin{aligned} \delta &= 1 - \beta X, \quad \Gamma = V - \beta Y, \quad \nu = \beta Z - T, \\ \tau &= Y - XV, \quad \Delta = Z - TX, \quad \mu = VZ - TY \end{aligned} \tag{B.3}$$

are strictly positive. In particular $P_1 = -1$ is built into the representation. The signs of g, h, v, j, q are not extra assumptions in this definition.

These inequalities retain the complete initial and successive Schur bounds. Eliminating the first two pivots gives the receiver/core blocks with pivots p, k; eliminating k gives $\left(\begin{smallmatrix} r & r \\ r & w \end{smallmatrix}\right)$. The last signed pivot is $w - r = -F$. In the positive high-value regime no division at a zero pivot is made. Sources with an earlier zero pivot have already been handled before entry to this chamber.

Appendix B.2. High-value budgets and simultaneous column reconstruction

The reductions in this subsection keep the seven-coordinate head $\eta = (t, \beta, V, T, X, Y, Z)$ fixed. Whenever the row scale E is fixed as well, the first, middle, and last columns always use the same E, p, k. The sign conclusions below are consequences of a complete source; they are not imposed on an enlarged independent-column relaxation.

Lemma Appendix B.2 (Elementary budgets of a complete high-value source)**.** *Every source of Definition Appendix B.1 with $F > 4$ satisfies*

$$2 < r \le k < 3, \quad 0 < g, j, h \le 1, \quad j < 1, \quad 1 < p < 1 + \beta < 2, \quad 0 < a < p, \tag{B.4}$$

$$0 < t \le 1, \quad X > \frac{1}{1+\beta} > \frac{1}{2}, \quad Ek < 1 - T, \quad r < \frac{5}{2}, \quad r + 2E + T < 3, \tag{B.5}$$

$$\Delta > \tau, \quad t > V, \quad Q_o := t\delta - \Gamma > 0, \quad \Omega := t\Delta + E\tau - \mu > 0,$$
$$\zeta := t\nu - E\Gamma > 0. \tag{B.6}$$

Moreover $q > 0$. If $F \ge 33/8$, the positive-receiver conclusion of Lemma Appendix B.4 also gives $v > 0$, and the quantitative bounds are

$$3 - r - 2E - T > \frac{1}{320}, \quad \Delta - \tau > \frac{1}{32}, \quad t - V > \frac{1}{48}, \quad q > \frac{41}{480}, \quad P_2 > \frac{61}{480}. \tag{B.7}$$

Proof. Since $|w| \le r$ and $D_{21} = r + Ea \le k$, we have $r \ge F/2 > 2$ and $k \ge r + Ea$. The identity $O_{00} = k + XP_0 - \delta g$, together with $P_0 \ge -1$, implies $g > 0$. If $j \le 0$, then $O_{00} = k - Xj - g \ge k - 1 > 1$, a contradiction. Similarly $O_{21} = r + Ea - Z - \nu h \le 1$ gives $h > 0$. Thus $|\widehat{q}_1| = 1 + \beta h \le p$ implies $p > 1$, and $P_0 = -j - \beta g \ge -1$ gives $j + \beta g \le 1$, in particular $j < 1$. Put $s = p - 1$ and $\epsilon = r - 2$. Initially $0 < s \le \beta \le 1$, and the same first column yields

$$k \le p + Xj, \qquad k \le 1 + Xj + g, \qquad k \le 1 + s + X(1 - \beta s).$$

For the last inequality substitute $j \le 1 - \beta g$: the function $\min(s, g) - \beta X g$ is maximized at $g = s$ on $[0, 1]$. Also $k \le 1 + X + (1 - \beta X)g \le 2 + (1 - \beta)X \le 3 - \beta$. Consequently $\beta < 1$, $p < 2$, and $0 < \epsilon \le s(1 - \beta) < s$. The middle-column original bound gives $r + Ea \le 1 + Z + \nu h \le 2 + \beta - T$, whence $T \le \beta - \epsilon$. Adding this bound for r to $r \le 3 - \beta$ proves $r < 5/2$.

The inequality $O_{11} = r - ta - Y + \Gamma h \le 1$ gives $ta > 0$, hence $t > 0$ and $a > 0$. The stage bound $S_{01} \ge -p$ gives $a + X(1 + \beta h) \le p$, so $a < p$. If $X \le 1/(1 + \beta)$, the preceding bound for k gives

$$k \le 1 + s + \frac{1 - \beta s}{1 + \beta} = 2 + \frac{s - \beta}{1 + \beta} \le 2,$$

a contradiction. In particular $X > \delta$. Since $Xj + g \ge k - 1 > 1$, we have $j + g > 1$. Now $Z > T$ and $O_{20} = -Ek - Zj - Tg \ge -1$ imply $Ek < 1 - T$. Thus $E < 1/2$. Finally,

$$(1 + \beta)Z - 1 - T = (r - 2) + Ea + (1 - O_{21}) + \nu(1 - h) > 0.$$

Together with $pZ \le 1 + T$, this proves the strict inequality $p \le (1 + T)/Z < 1 + \beta$.

Here is a positive-sum proof of the stronger row-scale budget. Set

$$A = r - p, \quad z_* = \frac{1 + \epsilon + Ts/\beta}{p}, \quad C_0 = 1 - Ek - \frac{Z}{X}(k - p) - Ts.$$

The exact identities

$$\begin{aligned} C_0 &= (1+O_{20}) + T(1-O_{00}) + \frac{\Delta}{X}(p - S_{00}) \geq 0, \\ B_R &:= 1 - Er - Z(r-p) - Ts \\ &= C_0 + (E+Z)(k-r) + Z(1/X - 1)(k-p) \geq 0, \\ \beta p(Z - z_*) &= \beta(Ea + 1 - O_{21}) + \nu(s - \beta h) \geq 0 \end{aligned}$$

show that $C := 1 - Er - z_*A - Ts = B_R + A(Z - z_*) \geq 0$. Define

$$\begin{aligned} m &= \frac{r}{2} - s - \frac{As}{\beta p} = \frac{(1-s)(2s-\epsilon)}{2p} + \frac{A(\beta - s)}{\beta p} > 0, \\ B &= 3s(1+s)(1-\beta) + 4(\beta - s) + (1+s)(\beta - s)(1-\beta), \\ N &= 2(\beta - s)\epsilon^2 + B\epsilon + 2\beta(1-\beta)(1-s^2) > 0. \end{aligned}$$

All displayed signs follow from $0 < \epsilon < s \leq \beta < 1$. Direct expansion, using $p = 1 + s$ and $r = 2 + \epsilon$, gives

$$W := 3 - r - 2E - T = \frac{N}{\beta p r} + \frac{2}{r}\{C + m(\beta - \epsilon - T)\} > 0.$$

This proves the row-scale budget without changing a matrix coordinate.

For $\Delta > \tau$, use $\Delta - \tau = \delta(Z - Y) + X(\Gamma + \nu)$. If $Y \geq Z$, then

$$h(\Delta - \tau) - X\epsilon = X\{Ea + 1 - Y + 1 - O_{21}\} + X\Gamma h + (X - \delta h)(Y - Z) \geq 0.$$

If $Y \leq Z$, the same difference equals

$$\delta h(Z - Y) + X\Gamma h + X\{Ea + 1 - Z + 1 - O_{21}\} \geq 0.$$

Thus $\Delta - \tau \geq X\epsilon/h > 0$. Let

$$\begin{aligned} B_0 &= (1+O_{20}) + T(1-O_{00}) \geq 0, \\ J &= \mu(1+O_{02}) + \Delta(1-O_{12}) + \tau(1+O_{22}) \geq 0. \end{aligned}$$

Expansion at the same complete source gives

$$\begin{aligned} \Delta b(t-V) - \frac{\Delta + \tau}{2}(F-4) &= \Delta(1-V) + \tau\{B_0 + \Delta j + (E+T)(k-b)\} \\ &\quad + J + \frac{\Delta - \tau}{2}(r+w) > 0. \end{aligned}$$

The strict sign follows already from $\tau\Delta j > 0$. Since $b \geq 0$, it implies $b > 0$ and $t > V$. Therefore $Q_o = \delta(t-V) + \beta\tau > 0$ and $\Omega = (t-V)\Delta + (E+T)\tau > 0$. For the last direction use

$$\zeta h - (t+E)(r-2) = E(1-Y) + t(1-Z) + E(1-O_{11}) + t(1-O_{21}) \geq 0.$$

The tail signs follow from two more slack identities:

$$\begin{aligned} 4 - W + (EX + Z)q - F &= (1+O_{22}) + E(1+O_{02}) \\ &\quad + (E+T)(1-v) \geq 0, \\ 4 - T - W + (EX+Z)P_2 - F &= (1+O_{22}) + E(1+O_{02}) + E(1-v) \\ &\quad + E\beta Xv + \nu v \geq 0 \quad (v \geq 0). \end{aligned}$$

The first, with $W > 0$, proves $q > 0$; the second proves $P_2 > 0$ after Lemma Appendix B.4 has supplied $v > 0$. That lemma uses only the preceding budgets, not this last positive-prefix conclusion.

For the quantitative assertions assume $F \geq 33/8$, so $\epsilon \geq 1/16$. Since $r \leq 3-\beta$, $1-\beta \geq \epsilon$, and $s \leq \beta$ gives $1-s \geq \epsilon$. Keeping only the constant term of N in the formula for W yields

$$W \geq \frac{2(1-\beta)(1-s)}{r} > \frac{2(1/16)^2}{5/2} = \frac{1}{320}.$$

The already proved direction estimates give $\Delta - \tau \geq X\epsilon/h > \epsilon/2 \geq 1/32$, and $t - V > (F-4)/(2b) > 1/48$, using $b \leq k < 3$. The middle and head bands also give

$$r + Ea \leq 1 + Z(1+\beta h) - Th \leq 1 + pZ - Th \leq 2 + T(1-h),$$

so $T > \epsilon \geq 1/16$. Finally $EX + Z < 3/2$, and the two tail slack identities imply

$$q > \frac{1/8 + 1/320}{3/2} = \frac{41}{480}, \qquad P_2 > \frac{1/8 + 1/16 + 1/320}{3/2} = \frac{61}{480}.$$

□

Lemma Appendix B.3 (The nonpositive-receiver tail face)**.** *On the complete canonical source of Definition Appendix B.1, the conditions $P_2 = 1$ and $v \leq 0$ imply*

$$r < \frac{33}{16}, \qquad F < \frac{33}{8}.$$

Proof. We give the analytic reduction and all the scalar polynomials used in its finite sign check. This lemma is restricted to the canonical $u_0 = 1$ source already under discussion; it does not need a further normalization of the first head coordinate.

Suppose instead that $r \geq R := 33/16$, and put $\epsilon = 1/16$, $\rho = 17/16 = R - 1$. Then $k \geq r + Ea \geq r > 2$. Directly from the first-column bands, $g > 0$ and $j > 0$: use $O_{00} = k + XP_0 - \delta g \leq 1$ with $P_0 \geq -1$ for the former; $j \leq 0$ would give $O_{00} = k - Xj - g \geq k - 1 > 1$. Also $O_{21} = r + Ea - Z - \nu h \leq 1$ implies $h > 0$, so $p \geq 1 + \beta h > 1$. Consequently $0 < s := p - 1 \leq \beta \leq 1$. Write $f = -v \in [0,1]$, so the tail-face equality gives $q = 1 + \beta f \geq 1$. If $t \leq 0$, then $O_{12} = r - tb + Yq - Vf \geq r + Y - V > 1$, a contradiction. Thus $0 < t \leq 1$. These facts use only $r > 2$, not the height conclusion being proved.

The first-column slack identity

$$\begin{aligned} V(p+Y) - \beta Y(p-1) - Vtk = \Gamma(p - S_{10}) + \beta Y(1 - O_{10}) \\ + VY(1 + P_0) \geq 0 \end{aligned}$$

retains the prefix bound at this same matrix. It gives

$$tk \leq 1 + s + Y - \frac{\beta Y s}{V}. \tag{B.8}$$

Let $d_0 = tX - Y$ be the physical head difference. The tail original bounds give

$$r \leq 1 + t + d_0 q + (V - t)f.$$

If $d_0 \leq 0$, then $q \geq 1$ and $0 \leq f \leq 1$ bound this expression by $1 + t + \max(0, V - t) \leq 2$. Hence $d_0 > 0$. Moreover $\tau = Y - XV > 0$ gives

$$t\delta - \Gamma = t - V - \beta d_0 > \frac{d_0 \delta}{X} > 0.$$

Substituting $q = 1 + \beta f$ therefore yields

$$r \leq 1 + t + d_0 - (t\delta - \Gamma)f \leq 1 + t + d_0, \quad d_0 \geq \rho - t > 0, \quad Y \leq t - d_0.$$

The middle-column receiver is denoted here by $\chi = -\widehat{q}_1 = 1 + \beta h$, to avoid confusing it with the first-column parameter j. Its two row-0 bounds give $a + X\chi \le \min(1 + s, 1 + h)$. Thus

$$r \le 1 + t + t\min(s, h) - d_0\chi - Vh. \tag{B.9}$$

From $L_2 \le 1$, $O_{21} \le 1$, and $Ea \ge 0$, we also have $r \le 1 + h + Z(\chi - ph)$. The factor $\chi - ph$ is positive, since otherwise $r \le 2$. Using $Z \le 1$, we obtain

$$\chi \ge \rho + sh, \qquad \epsilon \le (\beta - s)h, \qquad 0 < s \le \beta - \epsilon \le 15/16.$$

For the moment put $d = d_0$, and define

$$A = \rho(1 + d) - t > 0, \qquad B = ts - A.$$

Equation (B.9) implies both

$$A \le (t - V - ds)h, \qquad (V + ds)h \le B.$$

In particular $t - V - ds > 0$, $B > 0$. Comparing the two bounds on h, and using $A + B = ts$, gives

$$V \le \frac{B - ds^2}{s}, \qquad (\beta - s)B \ge \epsilon(V + ds).$$

Let

$$H = 1 + s + Y - tR, \quad J = B + \epsilon d > 0, \quad K = B - ds^2 > 0, \quad C = BH - \epsilon Y s.$$

From (B.8) and $k \ge R$, $VH \ge \beta Y s > 0$. Combining this with the preceding inequality gives $VC \ge Y s^2 J > 0$, hence $C > 0$. Now $V \le K/s$ and $V(Y + d) < tY$, the latter being the cofactor condition $V < Y/X$, imply

$$Q := HK - Ys(\epsilon + s^2) \ge 0, \qquad P := tC - s^2 J(Y + d) \ge 0. \tag{B.10}$$

For the first implication use the exact identity $KC - Ys^3 J = BQ$, with $B > 0$. For the second, $VC \ge Y s^2 J$ and $V < tY/(Y + d)$ even give $P > 0$; the weaker form above suffices.

We next lower the auxiliary difference to $d = \rho - t$. This is a relaxation of necessary inequalities, not a new physical matrix. The left side A decreases, $t - V - ds$ increases, $V + ds$ decreases, and B increases. The constraints $\epsilon \le (\beta - s)h$ and $VH \ge \beta Y s$ do not change; $V(Y + d) < tY$ and $Y \le t - d$ become weaker. Thus the same s, t, Y, V, β, h satisfies this relaxed system, and the preceding eliminations may be repeated at the new d. All letters A, B, H, J, K, C, Q, P henceforth use $d = \rho - t$. Since

$$K = t(1 + s + \rho + s^2) - \rho(1 + \rho + s^2) > 0,$$

we have

$$t > \ell(s) := \frac{17(16s^2 + 33)}{16(16s^2 + 16s + 33)}.$$

The coefficient of t is positive, and $t \le 1$, so $0 < K \le K|_{t=1} = s - 2\epsilon - \epsilon^2 - \epsilon s^2 < s - 1/8$. Every hypothetical source therefore supplies

$$\begin{aligned} \frac{1}{8} < s \le \frac{15}{16}, \qquad & \ell(s) < t \le 1, \\ 0 < Y \le Y_m := 2t - \rho, \qquad & H, K > 0, \quad Q, P \ge 0. \end{aligned} \tag{B.11}$$

Here is the complete finite exclusion of this scalar domain. Put $H_* = 1 + s - tR$ and $u_*(s) = 16(1 + s)/33$. Both Q and P are affine in Y. If $H_* \le 0$, write $Q = H_* K + Y\{K - s(\epsilon + s^2)\}$. Since $Q \ge 0$, its slope in Y must be nonnegative; therefore $Q \le Q(s, t, Y_m)$. The finite

sign checks below give $Q(s,t,Y_m) < 0$ throughout $[1/8, 15/16] \times [\ell(s), 1]$, a contradiction. If $H_* > 0$, then $t < u_*(s)$. The same checks give $u_* < \ell$ on $[1/8, 7/10]$, so $s > 7/10$. They also give $P(s,t,0) < 0$ and $P(s,t,Y_m) < 0$ on $[7/10, 15/16] \times [\ell(s), u_*(s)]$. Affineness in Y then contradicts $P \geq 0$ on the whole interval $[0, Y_m]$. This proves $r < R$, and $F \leq 2r < 33/8$.

To specify the finite checks without an external polynomial dictionary, write $f(s,t) = \sum_{i=0}^{n} f_i(s)t^i$, and define

$$\mathcal{B}_k[f; a_0, a_1, D_0] = \sum_{j=0}^{k} \frac{\binom{k}{j}}{\binom{n}{j}} (a_1 - a_0)^j \sum_{i=j}^{n} \binom{i}{j} f_i a_0^{i-j} D_0^{n-i}. \tag{B.12}$$

Here a_0, a_1, D_0 are polynomials in s. The polynomial $\mathcal{B}_k$ is D_0^n times the k-th Bernstein coefficient of f on $[a_0/D_0, a_1/D_0]$. Its formula divides by no interval length and remains valid at coincident endpoints. Let

$$\begin{aligned} D_s &= 16s^2 + 16s + 33, \qquad a = 17(16s^2 + 33), \\ d_Q &= 16D_s, \quad d_P = 528D_s, \quad b_P = 256(1+s)D_s. \end{aligned}$$

All these denominators are strictly positive for $s \geq 0$. The eleven coefficient polynomials and the twelfth polynomial are specified exactly by

$$\begin{aligned} Z_{Qi} &= -\mathcal{B}_i[Q(s,t,2t-\rho); a, d_Q, d_Q], && 0 \leq i \leq 2, \\ Z_{0i} &= -\mathcal{B}_i[P(s,t,0); 33a, b_P, d_P], && 0 \leq i \leq 3, \\ Z_{mi} &= -\mathcal{B}_i[P(s,t,2t-\rho); 33a, b_P, d_P], && 0 \leq i \leq 3, \\ Z_u &= 33a - b_P. \end{aligned}$$

The degrees in t used in (B.12) are respectively 2, 3, and 3. Also $u_* - \ell = -Z_u/d_P$. For any univariate polynomial $z(s) = \sum_{i=0}^{N} z_i s^i$, its Bernstein coefficients on $[L, H]$ are explicitly

$$b_k = \sum_{j=0}^{k} \frac{\binom{k}{j}}{\binom{N}{j}} (H-L)^j \sum_{i=j}^{N} z_i \binom{i}{j} L^{i-j} \qquad (0 \leq k \leq N). \tag{B.13}$$

Applying this rational formula to the twelve displayed polynomials gives the following thirteen checks. In every row, every coefficient is strictly greater than 1 under the stated common-denominator normalization.

Polynomial	Degree	Interval
Z_{Q0}, Z_{Q1}	7	$[1/8, 15/16]$, each
Z_{Q2}	7	$[1/8, 17/32]$ and $[17/32, 15/16]$
Z_{00}	7	$[7/10, 15/16]$
Z_{01}	10	$[7/10, 15/16]$
Z_{02}, Z_{03}	11	$[7/10, 15/16]$, each
Z_{m0}	9	$[7/10, 15/16]$
Z_{m1}	10	$[7/10, 15/16]$
Z_{m2}, Z_{m3}	11	$[7/10, 15/16]$, each
Z_u	3	$[1/8, 7/10]$

These inequalities are fixed finite rational arithmetic statements: the formulas above define every coefficient to be checked. Positive Bernstein coefficients imply strict positivity because the Bernstein basis is nonnegative and sums to 1 on the closed unit interval. Thus $Z_{Qi}, Z_{0i}, Z_{mi} > 0$ give the required strict negative t-coefficients, and $Z_u > 0$ gives the endpoint ordering. Empty t-intervals impose no condition; degenerate ones are covered by the same formulas. The two intervals for Z_{Q2} cover the whole required s-range. This is a fixed thirteen-check certificate, with no adaptive subdivision or unreported numerical optimization. $\square$

Lemma Appendix B.4 (Positive tail receiver)**.** *On the complete canonical source, $v \le 0$ implies $F < 33/8$. The required tail-face bound is Lemma Appendix B.3, whose fixed polynomial-positivity certificate is displayed above.*

Proof. Only $F > 4$ needs consideration. If $v \le 0$, then $O_{02} = -b + XP_2 + \delta v \ge -1$ gives $b \le 2 < k$. The preceding budgets give $q > 0$, $t > V$, and $r + E < 3$. From $O_{12} - tO_{02} = r - (tX - Y)q + (V - t)v$ we obtain $r \le 1 + t + (tX - Y)q + (t - V)v$; thus $tX - Y > 0$. Increase v, b together by

$$0 \le \lambda \le \min\{-v, (1 - P_2)/\beta\}.$$

This fixes p, r, w, F, the head, E, and both other columns. The changes in the last original column are $(0, -(t - V)\lambda, (E + T)\lambda)$, while P_2 increases by $\beta\lambda$. Throughout the segment $v' \le 0$, $P_2' \le 1$, and $0 \le b' \le 2 < k$. The original lower bound in row 1 is safe because $O_{12}' \ge r-2-V > -1$; its upper bound improves. The stage bounds satisfy $S_{02}' = O_{02} - v' \ge -1$ and $S_{12}' = r - tb' + Yq > 0$, while their upper bounds improve. Both row-2 lower bounds improve, and

$$S_{22}' = w + Eb' + Zq \le w + Ek + Zp < w + 2 < 1, \qquad O_{22}' \le S_{22}'.$$

Here $Ek < 1 - T$, $pZ \le 1 + T$, and $w + 2 = r - F + 2 < 1/2$. The core entry $r - tb'$ stays positive; $w + Eb'$ lies between $w \ge -k$ and $S_{22}' < k$. All remaining changed bounds are either fixed or improved. Hence the whole segment is physical.

Its endpoint has $P_2' = 1$ or $v' = 0$. In the latter case, if $P_2' < 1$, then $0 < q = P_2' < 1$. Increase q by $0 \le \sigma \le 1 - q$, increase b' by $X\sigma$, and keep $v' = 0$. The original changes are $(0, -(tX - Y)\sigma, (EX + Z)\sigma)$. We still have $b'' \le 1 + Xq'' \le 2$; now $O_{12}'' \ge r - 2 > 0$, $O_{22}'' < w + 2 < 1$, and $S_{i2}'' = O_{i2}''$. The same core estimates apply. This second physical segment ends at $P_2'' = 1, v'' = 0$. The finite safe-tail exclusion therefore applies to a complete source with the original r, F, proving the claim. No rank assumption is added and no cofactor changes in either segment. □

Lemma Appendix B.5 (Normalization of the second pivot)**.** *Every complete source with $F > 4$ can be replaced, at the same head, E, D, F, by one with $p = \bar{p} := (1 + T)/Z$. This replacement preserves the entire original tail block O and all prefix coordinates P_j.*

Proof. The preceding proof gives $p \le \bar{p} < 1 + \beta < 2$. Increase only p along $[p, \bar{p}]$. The $\widehat{q}, S$ bounds are relaxed. All tail and core coordinates are fixed. For the head column, its lower bounds follow from $p'x_i - u_i \ge -u_i \ge -1$; its upper bounds follow from

$$p'X < 2, \qquad p'Y < (1 + \beta)Y < 1 + V, \qquad p'Z \le 1 + T.$$

The middle strict inequality uses $1 + V - (1 + \beta)Y = 1 - Y + \Gamma > 0$. Also $0 < p' - \beta < 1$. Thus every point of the segment is a complete physical source, and at its endpoint $L_2 = 1$. □

Lemma Appendix B.6 (The complete twelve first-column capacities)**.** *Fix $0 < \beta, V, T, X, Y, Z \le 1$, all six positive cofactors, $0 \le E \le 1$, $V < t \le 1$, and $1 < p = \bar{p} < 1 + \beta < 2$. These assumptions hold for a complete high-value head by Lemmas Appendix B.2 and Appendix B.5; the recovery assertion does not require an already completed middle or last column. Define*

$$\begin{aligned} j_L(k) &= \max\{(k - p)/X, (tk - p)/Y, (tk - 1 - V)/Y\}, \\ j_U(k) &= \min\{(1 + \beta - \beta k)/\delta, (1 + T - (E + T)k)/\Delta, \\ &\qquad (V + \beta - \beta tk)/\Gamma, (V + T - (VE + Tt)k)/\mu\}. \end{aligned}$$

For $k > 2$, a complete first column exists if and only if $j_L(k) \le j_U(k)$. A single witnessing pair is

$$j = j_L(k), \qquad g = \max\{k - 1 - Xj, (tk - 1 - Yj)/V\}. \tag{B.14}$$

Write the three lower affine functions as $\lambda_i k - \eta_i$ and the four upper functions as $\xi_j - \kappa_j k$, where

$$(\lambda_i, \eta_i) = (1/X, p/X),\ (t/Y, p/Y),\ (t/Y, (1+V)/Y),$$
$$(\xi_j, \kappa_j) = \begin{array}{c} ((1+\beta)/\delta, \beta/\delta),\ ((1+T)/\Delta, (E+T)/\Delta), \\ ((V+\beta)/\Gamma, \beta t/\Gamma),\ ((V+T)/\mu, (VE+Tt)/\mu). \end{array}$$

Then the complete first-column capacity is

$$K = \min_{1\le i\le 3,\, 1\le j\le 4} K_{ij}, \qquad K_{ij} := \frac{\eta_i + \xi_j}{\lambda_i + \kappa_j}. \tag{B.15}$$

Every denominator is positive. Raising an existing $k > 2$ to K, while reconstructing the first column, preserves the head, E, p, r, w, F and the other two columns.

Proof. The first-column constraints require

$$j \ge (k-p)/X, \quad j \ge (tk-p)/Y,$$
$$g \ge \max\{k-1-Xj, (tk-1-Yj)/V\},$$
$$g \le \min\{1, (1-j)/\beta, (1-Ek-Zj)/T\}.$$

Comparing the two lower bounds for g with its last two upper bounds gives exactly the four terms of j_U. The first lower bound for g is at most $p-1 < 1$; the second is at most 1 precisely when $j \ge (tk-1-V)/Y$. This proves necessity and gives (B.14) when the intervals intersect.

For completeness, the omitted physical bounds follow from this same pair. We have $j > 0$, and the first upper bound in j_U gives $j < 1$, since $\beta(1+X-k) < 0$. Then $g \ge k-1-Xj > k-2 > 0$, while the interval comparisons give $g \le 1$. Thus $O_{00} = k - Xj - g > k-2 > 0$ and $O_{10} = tk - Yj - Vg > t(k-g) - 1 > -1$, using $t > V$. Their upper bounds have already been imposed. The stage entries S_{00}, S_{10} lie above -1 and below p. Also $-1 \le O_{20} < 0$ gives $-1 \le S_{20} < 0$; $P_0 = -j - \beta g \in [-1, 0)$, and the receiver bounds follow from $0 < j < 1 < p$, $0 < g \le 1$. Finally the first core column obeys $|D_{i0}| \le k$ because $0 < t \le 1$, $0 \le E \le 1$. This is a complete column.

Every lower–upper comparison is $(\lambda_i + \kappa_j)k \le \eta_i + \xi_j$, proving (B.15). At $k = K > 2$ the displayed construction is therefore feasible. Interpolate linearly between the old and new triples (k, j, g): all first-column conditions are affine at the fixed head and E, p, while increasing k relaxes all other core bounds and $a, b \le k$. The other two columns and the height stay fixed. The reconstructed first column itself gives $k \le 1 + Xj + g \le 2 + (1-\beta)X < 3$ and $j + g > 1$; together with $Z > T$ and $Ek + Zj + Tg \le 1$, this proves $Ek < 1 - T$ without assuming the existence of either of the other columns. □

Lemma Appendix B.7 (Strict gaps at an actually completed high-value capacity)**.** *Every complete source with $F > 4$ satisfies*

$$O_{00} < 3 - F/2, \qquad g < 3 - F/2, \qquad F + \beta < 5.$$

Consequently, at a complete high-value source with $k = K$, all eight capacities other than $K_{13}, K_{14}, K_{23}, K_{24}$ are strictly larger than K. In particular

$$K_{13} = \frac{p\Gamma + X(V+\beta)}{\Gamma + \beta t X}, \qquad K_{23} = \frac{p\Gamma + Y(V+\beta)}{tV},$$
$$K_{14} = \frac{p\mu + X(V+T)}{\mu + X(VE+Tt)}, \qquad K_{24} = \frac{p\mu + Y(V+T)}{V(tZ+EY)}.$$

These four capacities may replace the twelve at this completed source and in any neighborhood where the strict gaps persist. This statement does not assert that deleting eight inequalities from an arbitrary uncompleted or low-value first-column problem is harmless.

Proof. With the nonnegative tail slack J used above,

$$\mu(1-O_{00}) = \tau(F-4) + (\Delta-\tau)(r-2) + \Omega(k-b) \\ + \Delta(1-O_{10}) + \tau(1+O_{20}) + J.$$

Since $r-2 \geq (F-4)/2$ and $0 < \mu = V\Delta - T\tau < \Delta + \tau$, this yields $1 - O_{00} > (F-4)/2$. Furthermore

$$\beta X(1-g) = (k-2) + (p - S_{00}) + X(1+P_0) \\ + (1+\beta-p) + (1-\beta)(1-X) \geq k-2,$$

and $0 < \beta X < 1$ gives $1 - g > k - 2 \geq F/2 - 2$. Finally

$$O_{00} - (k - 2 + \beta) = (1-X)j + (1+P_0) + (1-\beta)(1-g) > 0,$$

so $k \geq F/2$ proves $F + \beta < 5$.

If a first-column capacity is attained, its lower and upper affine bounds for j coincide and equal the actual j. For columns 1 or 2 of the capacity array, equality forces the O_{00} lower bound on g to equal a prefix or row-2 upper bound, hence $O_{00} = 1$, which is impossible. For row 3 of the capacity array, equality forces the O_{10} lower bound for g to be 1, hence $g = 1$, also impossible. The excluded indices are precisely the complement of $\{13, 14, 23, 24\}$. All inequalities are strict at the complete source, so continuity preserves them locally. □

Lemma Appendix B.8 (Recovering the omitted first-column bounds from a common tail)**.** *Fix $0 < \beta, V, T, X, Y, Z \leq 1$, all six positive cofactors, $0 \leq E \leq 1$, and $V < t \leq 1$. Assume $\Delta > \tau$, $\Omega > 0$, and $1 < p = \bar{p} < 1 + \beta < 2$. Suppose*

$$2 < k \leq K_{13}, K_{14}, K_{23}, K_{24}, \qquad 2 < r \leq k, \quad |w| \leq r, \quad F = r - w > 4,$$

and suppose a tail choice $0 \leq b \leq k$ has

$$J_{\text{tail}} := \mu + \Delta + \tau - \Delta r + \tau w + \Omega b \geq 0. \tag{B.16}$$

Then the first column has the complete physical reconstruction

$$j = \max\{(k-p)/X, (tk-p)/Y\}, \qquad g = (tk - 1 - Yj)/V. \tag{B.17}$$

In particular $O_{10} = 1$, and the original twelve-capacity test holds. If $k = \min\{K_{13}, K_{14}, K_{23}, K_{24}\}$, then $k = K$ as defined in (B.15)*.*

Proof. The four capacity inequalities say that each of the two lower bounds in (B.17) is at most each of the last two upper bounds in j_U. Substitution therefore gives

$$S_{00} \leq p, \quad S_{10} \leq p, \quad O_{10} = 1, \quad j + \beta g \leq 1, \quad Ek + Zj + Tg \leq 1.$$

At this point no sign or receiver bound for g is presumed. The first-column gap identity used in Lemma Appendix B.7, being algebraic, now gives

$$\mu(1-O_{00}) = \tau(F-4) + (\Delta-\tau)(r-2) + \Omega(k-b) + \tau(1+O_{20}) + J_{\text{tail}} > 0.$$

Thus $O_{00} < 1$ follows from the common tail slack, without assuming that the unfinished first column is already physical. We have $j > 0$ because $k > 2 > p$. If $g \leq 0$, then $j + \beta g \leq 1$ implies $O_{00} = k - Xj - g \geq k - X - \delta g \geq k - 1 > 1$, a contradiction. Hence $g > 0$ and $j < 1$. Combining $k - Xj \leq p$ with $j \leq 1 - \beta g$ gives

$$\beta X g \leq p + X - k < \beta + X - 1 \leq \beta X,$$

so $g < 1$. Now $O_{00} > k - 2 > 0$, $O_{10} = 1$, and $-1 \le O_{20} < 0$. The stage entries satisfy $S_{00} > k - 1 > 0$, $S_{10} = 1 + Vg > 0$, and $-1 \le S_{20} < 0$; their upper bounds were already imposed. The prefix, receiver, and first-core-column bounds follow as in Lemma Appendix B.6. This is a complete first column. Consequently $k \le K$. Conversely K is the minimum of all twelve capacities and hence at most their four-term minimum, which proves the final assertion. The argument makes no claim about a four-capacity point lacking the common positive-height tail slack. □

Lemma Appendix B.9 (The complete middle-column capacity)**.** *Fix head coordinates in* $(0, 1]$ *with all six cofactors positive. Assume*

$$0 \le E \le 1, \quad V < t \le 1, \quad X > \delta, \quad \zeta = t\nu - E\Gamma > 0,$$
$$p = \bar{p}, \qquad 1 < p < 1 + \beta < 2.$$

Fix $k > 2$ *for which a complete first column has been supplied; in particular one may use Lemma Appendix B.8 or* $k = K$ *above. No pre-existing complete high-value matrix is assumed in the converse. Put* $H = (p-1)/\beta \in (0, 1)$*, and define*

$$A_1 = 1 + Y + t(1 - X), \qquad B_1 = Q_o,$$
$$A_2 = \frac{t(1+Z) + E(1+Y)}{t+E}, \qquad B_2 = \frac{\zeta}{t+E},$$
$$C_1 = 1 + Y + t(p - X), \qquad D_1 = \Gamma + t\beta X,$$
$$C_2 = \frac{tk + E(1+Y)}{t+E}, \qquad D_2 = \frac{E\Gamma}{t+E}.$$

Then $B_1, B_2, D_1 > 0$*,* $D_2 \ge 0$*, and the middle column can carry a prescribed* $r > 2$ *if and only if*

$$r \le \mathcal{R}_{\text{mid}} := \min \left\{ \begin{matrix} C_1, C_2, A_1 + B_1 H, A_2 + B_2 H, \\ \dfrac{B_i C_j + D_j A_i}{B_i + D_j} \quad (1 \le i, j \le 2) \end{matrix} \right\}. \tag{B.18}$$

For every such r*, the single reconstruction is*

$$h = \max\{0, (r - A_1)/B_1, (r - A_2)/B_2\},$$
$$a = \frac{r - 1 - Y + \Gamma h}{t}, \qquad \widehat{q}_1 = -1 - \beta h. \tag{B.19}$$

It has $O_{11} = 1$ *and* $O_{01} = -1$ *or* $O_{21} = 1$*. It uses exactly the* p, k, E *of the reconstructed first column.*

Proof. First consider an existing middle column. Lower a to $a_* = (r - 1 - Y + \Gamma h)/t > 0$. Since $O_{11} \le 1$, this does not increase a; it raises O_{11} to 1. The potentially increasing O_{01} remains below zero because $O_{01} = -a - X + \delta h < \delta - X < 0$. The stage entry S_{01} rises but stays negative. Both O_{21} and D_{21} decrease and remain positive, because $r > 2$, $Z \le 1$, and $\nu h < 1$. At the endpoint $D_{11} = 1 + Y - \Gamma h \in (0, 2)$ and $S_{11} = 1 - Vh \in (0, 1]$; their lower bounds improve and their upper bounds hold throughout. Thus the segment is physical and fixes every other column and r, w.

With $O_{11} = 1$, substitution of $a = a_*$ gives exactly

$$t(1 + O_{01}) = A_1 + B_1 h - r,$$
$$t(1 - O_{21}) = (t + E)(A_2 + B_2 h - r),$$
$$t(p + S_{01}) = C_1 - D_1 h - r,$$
$$t(k - r - Ea) = (t + E)(C_2 - D_2 h - r).$$

Together with $0 \le h \le H$, these are the four line tests $r \le A_i + B_i h, C_j - D_j h$. Their interval intersection is nonempty exactly under the eight comparisons in (B.18). This remains valid when $E = 0, D_2 = 0$, because every $B_i + D_j$ stays positive. Its smallest admissible h is (B.19). Since $A_2 \le 2 < r$, this h is positive and at least one of the two increasing lines is attained.

Conversely use this h, a. The four identities impose the two original bounds, the stage bound, and $D_{21} \le k$. We have $a > 0$, while the stage bound gives $a + X(1 + \beta h) \le p$, hence $a < p < 2 < k$. The other core entry is $D_{11} = 1 + Y - \Gamma h \in (0, 2)$. The stage entries are $S_{11} = 1 - Vh \in (0, 1]$ and $S_{21} = r + Ea - Z(1 + \beta h) \ge r - p > 0$; also $S_{21} \le O_{21} \le 1$. The remaining original bounds follow from $O_{01} < \delta - X < 0$ and $O_{21} > r - 2 > 0$. Receiver and prefix bounds follow from $0 < h \le H < 1$, $1 + \beta h \le p$, and $P_1 = -1$. All head quantities, the first column, and the last column remain unchanged. This proves simultaneous physical reconstruction, rather than independent existence claims for different columns. □

Appendix B.3. The height polygon and simultaneous wall reconstruction

Throughout this subsection put $H_0 = 33/8$, and fix the seven-coordinate head $\eta = (t, \beta, V, T, X, Y, Z)$. Cofactors always have the meanings in (B.3). We use Γ, not the height threshold γ, for the cofactor $V - \beta Y$. Define

$$Q_o = t\delta - \Gamma, \quad \ell = \nu - E\delta, \quad \Omega = t\Delta + E\tau - \mu, \quad \zeta = t\nu - E\Gamma.$$

The common-column results above give, in the high-value domain, $t > V$, $\Delta > \tau$, $Q_o > 0$, $\Omega > 0$, $p = (1 + T)/Z < 1 + \beta < 2$, and $X > 1/(1 + \beta)$. They also provide one first column at the complete twelve-capacity minimum $k = K(E)$, and one middle column for every $2 < r \le \mathcal{R}_{\mathrm{mid}}(E)$. These are simultaneous completions at the same η, E, p, k.

Lemma Appendix B.10 (The positive tail slope gap)**.** *A complete canonical source with $F \ge H_0$ has $\ell > 0$.*

Proof. Suppose instead that $\ell \le 0$. Write $\epsilon = r - 2 \ge 1/16$, $s = p - 1$, and, for this proof only, $A = (1 + E)\epsilon$, $B = \beta$, $D_B = 1 - B^2$. The common-column budgets give $0 < s \le B < 1$, $E < 1/2$, and $k \le 1 + s + X(1 - Bs)$. The middle-column identity

$$t(a - \epsilon) = \epsilon(1 - t) + (1 - Y) + \Gamma h + (1 - O_{11}) > 0$$

gives $a > \epsilon$. Since $\nu > 0$ and $\nu \le E\delta$, also $E > 0$, and hence $k \ge r + Ea > 2 + A$. It follows that

$$A \le s(1 - B) \le B(1 - B), \qquad X \ge \frac{1 + A - s}{1 - Bs}.$$

The difference between the last lower bound and its value at $s = B$ is

$$\frac{(B - s)(1 - B - BA)}{(1 - Bs)(1 - B^2)} \ge 0,$$

where $1 - B - BA \ge (1 - B)(1 - B^2) > 0$. Consequently

$$0 < \delta \le \bar{\delta} := \frac{1 - B - BA}{1 - B^2}.$$

Two further same-matrix identities are

$$\begin{aligned} E\delta - (1 + A - Z) &= (E\delta - \nu) + \nu(1 - h) \\ &\quad + E(a - \epsilon) + (1 - O_{21}) > 0, \\ 1 - Ek - \frac{Z}{X}(k - p) - Ts &= (1 + O_{20}) + T(1 - O_{00}) + \frac{\Delta}{X}(p - S_{00}) \ge 0. \end{aligned}$$

Using $X \le 1$, $k > 2 + A$, $s \le B$, $Z > T$, and $T = BZ - \nu \ge BZ - E\delta$, they imply

$$E\bar{\delta} > A, \qquad 1 > E(2 + A - B\bar{\delta}) + (1 + A - E\bar{\delta})(1 + A - B + B^2).$$

The substitution $\delta \le \bar{\delta}$ in the second inequality has the stated direction: its right-hand side before substitution has coefficient $-E(1 + A + B^2) < 0$ in δ.

Set

$$\begin{aligned}\mathfrak{F}(A, E, B) &= (1 - B^2)\{E(2 + A - B\bar{\delta}) \\ &\qquad + (1 + A - E\bar{\delta})(1 + A - B + B^2) - 1\}, \\ \mathfrak{G}(A, E, B) &= E(1 - B) - (1 - B^2 + EB)A.\end{aligned}$$

Thus $\mathfrak{F} < 0 < \mathfrak{G}$. Direct expansion gives

$$\mathfrak{F}_A = (1 - B^2 + EB)(2A + B^2 - B + 2) > 0, \qquad \mathfrak{G}_A = -(1 - B^2 + EB) < 0.$$

Replacing A by $A_0 = (1+E)/16$ preserves these necessary inequalities. Put $C = B^2 - 17B + 15$, $D = 1 - B^2$. Then $16\mathfrak{G}(A_0, E, B) = CE - BE^2 - D > 0$, so $C > 0$, $E > D/C$, and $C > 2D$. Since $3B^2 - 17B + 13$ decreases on $[0, 1]$ and is negative at $11/12$, we obtain $0 < B < 11/12$.

Writing $\mathfrak{F}_0(E, B) = \mathfrak{F}(A_0, E, B)$, expansion gives

$$256(\mathfrak{F}_0)_E = 3BE^2 + (32B^3 - 34B^2 + 68B + 2)E + H_4(B),$$

where

$$H_4 = -16B^4 + 288B^3 - 802B^2 + 273B + 290.$$

The Bernstein coefficients of H_4 on $[0, 1]$ are $290, 1433/4, 1757/6, 663/4, 33$, all positive. Therefore $\mathfrak{F}_0$ increases in $E \ge 0$. Moreover

$$\begin{aligned}\mathfrak{F}_0(D/C, B) &= \frac{(1 - B)^2(1 + B)P_7(B)}{256C^3}, \\ P_7(B) = &- 256B^7 + 13072B^6 - 233488B^5 + 1641728B^4 \\ &- 3447057B^3 + 3056607B^2 - 1206016B + 176640.\end{aligned}$$

On the three intervals $[0, 3/8]$, $[3/8, 1/2]$, and $[1/2, 11/12]$, its degree-seven Bernstein coefficients have respective positive minima

$$\frac{52615035}{16384}, \qquad \frac{31756901}{17920}, \qquad \frac{43980727}{559872}.$$

These three finite coefficient calculations can be recovered directly from the displayed polynomial: after substituting $B = a + (b - a)z$, if the power coefficients are c_j, the Bernstein coefficient of index i is $\sum_{j=0}^{i} c_j \binom{i}{j} / \binom{7}{j}$. Thus $P_7 > 0$ on the entire required interval, and $\mathfrak{F}_0(E, B) > \mathfrak{F}_0(D/C, B) > 0$, a contradiction. This proof uses four fixed univariate positive-coefficient certificates; it does not use an adaptive multidimensional tree. □

For the reconstruction statements below, the head satisfies the common-column entrance inequalities just listed, and $0 \le E < \min\{1, \nu/\delta\}$; in particular $\ell > 0$. The quantities K and $\mathcal{R}_{\mathrm{mid}}$ are the complete capacities, with $K > 2$ and $\mathcal{R}_{\mathrm{mid}} > 2$. When a reconstruction is applied at $r > 2$, its first and middle columns are supplied by those capacity lemmas before the tail is constructed. Put

$$\begin{gathered}b_0 = 1 + X, \qquad a_T = 1 - Y + \Gamma b_0/\delta, \quad c_T = 1 + Z + \nu b_0/\delta, \\ R_T(b) = a_T + Q_o b/\delta, \quad C_T(b) = c_T - \ell b/\delta, \quad C_0 = C_T(b_0) = 1 + Z + Eb_0, \\ A_c = \mu + \Delta + \tau, \qquad B_c = \Delta + \nu + \delta, \\ \mathcal{C} = A_c + \Omega K, \qquad \mathcal{D} = \ell a_T + Q_o c_T.\end{gathered} \tag{B.20}$$

Here R_T, C_T are scalar tail capacities, not matrices. Two useful cofactor identities are

$$\Delta Q_o - \tau\ell = \Omega\delta, \qquad \Delta\mathcal{D} = \ell A_c + \Omega B_c.$$

Lemma Appendix B.11 (Anchor bounds and lowering the shared coordinate)**.** *One has $a_T < 2 < c_T$. At fixed head, decreasing E preserves a completed first and middle column at any $r > 2$. After replacing the first column by its new complete maximum $K(E)$, the same r remains feasible. In particular $\mathcal{R}_{\mathrm{mid}}(E)$ is nonincreasing on its high-value range, in the precise sense that*

$$E_1 \le E_2, \quad 2 < r \le \mathcal{R}_{\mathrm{mid}}(E_2) \quad \Longrightarrow \quad r \le \mathcal{R}_{\mathrm{mid}}(E_1).$$

This statement concerns the first two columns only.

Proof. From $Q_o > 0$, $t \le 1$, and $\tau > 0$, one obtains $\Gamma < \delta$ and $\Gamma X < \delta Y$. If $Y \ge X$, then $a_T < 2 + X - Y \le 2$; if $Y \le X$, then $a_T < 1 + Y/X \le 2$. Also $p < 1 + \beta$ gives $\nu + Z > 1$, whence $c_T > 1 + Z + \nu > 2$.

For the monotonicity assertion retain all completed variables except E. In the first column the only affected entries are $-Ek$, $-Ek - Zj$, and $-Ek - Zj - Tg$. They increase toward zero when E decreases, and remain negative because $j, g > 0$; all their absolute-value bounds are preserved. In the middle column the affected entries are $r + Ea$, $r + Ea - Z(1 + \beta h)$, and $r + Ea - Z - \nu h$. Their upper bounds improve. Their lower bounds remain valid since

$$r + Ea - Z(1 + \beta h) \ge r - Zp = r - 1 - T > 0,$$

and the last original entry is this stage entry plus $Th \ge 0$. Here $1 + \beta h \le p$, $pZ = 1 + T$, and $r > 2$. The core entry $r + Ea$ remains positive. These are all occurrences of E in the two columns. Finally each original first-column capacity has a positive constant numerator and a positive affine denominator with nonnegative slope in E; hence K cannot decrease when E is lowered. Enlarging the permitted core bound from the old k to this new K, and using the common-column reconstruction, preserves the middle column. No assertion about preserving the tail column under this operation has been used. □

Lemma Appendix B.12 (Tail slacks at one complete matrix)**.** *Write $x_r = r$, $y = -w$, and define*

$$J_a = \mu(1 + O_{02}) + \Delta(1 - O_{12}) + \tau(1 + O_{22}),$$
$$J_b = \Delta(1 - P_2) + \nu(1 + O_{02}) + \delta(1 + O_{22}).$$

Then the following exact identities hold:

$$\begin{aligned} J_a &= A_c + \Omega b - \Delta x_r - \tau y, \\ J_b &= B_c - \ell b - \delta y, \\ \ell J_a + \Omega J_b &= \Delta(\mathcal{D} - \ell x_r - Q_o y), \\ C_0 - y &= (1 + O_{22}) + E(1 + O_{02}) + (EX + Z)(1 - P_2) + \ell v. \end{aligned} \tag{B.21}$$

At a complete source of height at least H_0, all displayed slacks are nonnegative, and $C_0 - y > 0$.

Proof. Expand $O = S + u\widehat{v}^T$, $S = D + x\widehat{q}^T$, and $P = \beta\widehat{v} + \widehat{q}$, and cancel terms by the six cofactor definitions. Every weight in J_a, J_b is positive. In the last identity all terms are nonnegative and $\ell v > 0$, by Lemmas Appendix B.10 and Appendix B.4. □

The height polygon is the following closed set of scalar pairs:

$$\begin{aligned} 0 \le y \le x_r \le \mathcal{R}_{\mathrm{mid}}, \qquad & y \le C_0, \\ \Delta x_r + \tau y \le \mathcal{C}, \qquad & \ell x_r + Q_o y \le \mathcal{D}. \end{aligned} \tag{B.22}$$

Every complete high-value source maps to this polygon after normalizing p, k: the first oblique inequality follows from $b \le K$ and $J_a \ge 0$, the second from their positive combination in (B.21), and $y \le C_0$ from the last identity. Also $y = F - r > 0$, because $r < 5/2 < F$. This argument does not assume that the original matrix already has any of the three tail contacts.

Lemma Appendix B.13 (Complete recovery on the wall)**.** *Fix normalized common first and middle columns, and suppose $r \ge H_0/2$. There is a complete tail at $w = -r$ if and only if*

$$r \le \mathcal{R}_{\mathrm{mid}}(E), \qquad r \le \mathcal{W}(E), \qquad \mathcal{W}(E) := \min\left\{C_0, \frac{\mathcal{C}}{\Delta+\tau}, \frac{\mathcal{D}}{\ell+Q_o}\right\}.$$

The sufficiency assertion uses the complete twelve-capacity first column. It does not treat four selected capacities as a standalone relaxation. If a complete wall of height $m \ge f > H_0$ exists at this head and E, another complete wall of height exactly f exists there, with all columns reconstructed at common p, K, E.

Proof. Necessity is (B.22) at $x_r = y = r$. For sufficiency first suppose $r \le R_T(K)$. Set

$$b = \max\{b_0, \delta(r - a_T)/Q_o\}, \qquad v = (b - b_0)/\delta, \qquad q = 1 - \beta v.$$

Then $b_0 \le b \le K$, $R_T(b) \ge r$, and $C_T(b) \ge r$. For the last inequality use $r \le C_0$ when $b = b_0$, and $(\ell + Q_o)r \le \mathcal{D}$ otherwise. The capacity $K \le b_0 + \delta(p-1)$ gives $0 \le v \le p - 1 < 1$; hence $0 < q \le 1$, $P_2 = 1$, and $O_{02} = -1$. The two remaining original entries obey $O_{12} = 1 + r - R_T(b) \le 1$ and $O_{22} = -r + C_T(b) - 1 \ge -1$. Their opposite bounds follow from

$$O_{12} > r - K > -1, \qquad O_{22} \le -r + EK + Zq + Tv < 2 - r < 0.$$

The strict upper estimate uses $EK < 1 - T$, $Zq \le 1$, and $Tv \le T$. For the stage bounds, $S_{02} = -1 - v \ge -p$, $S_{12} > r - K > -1 > -p$, and $S_{22} = O_{22} - Tv \ge -1 - Tv \ge -p$. All stage upper bounds follow from $S_{i2} = O_{i2} - u_i v \le 1 < p$. The core entries satisfy $r - tb \ge r - K > -K$, $-r + Eb \ge -r \ge -K$; their upper bounds follow from the corresponding original entries after subtracting the nonnegative completion terms. Also $b \le K$. These verify every tail band in (B.2).

If instead $r > R_T(K)$, set

$$b = K, \qquad q = \frac{1 + V + (t - V)K - r}{\tau},$$
$$v = \frac{Xr - (tX - Y)K - X - Y}{\tau}.$$

Then $O_{02} = -1$, $O_{12} = 1$, and $O_{22} \ge -1$ is exactly $(\Delta + \tau)r \le \mathcal{C}$. At $r = R_T(K)$ these formulas agree with the preceding formulas at $b = K$; increasing r increases v and decreases P_2. Thus $v > 0$ and $P_2 < 1$. The missing upper bound for v must not be assumed. It follows from the independent identity

$$\begin{aligned}\mu(p + S_{02}) = {} & X(V + T)(r - 2) \\ & + [X(VE + Tt) + \mu](K - b) + \mu(p - S_{00}) \\ & + XV[(1 + O_{20}) + (1 + O_{22})] \\ & + XT[(1 - O_{10}) + (1 - O_{12})] > 0.\end{aligned}$$

All quantities on its right have already been verified, and its first term is strictly positive. Since $S_{02} = -1 - v$, it gives $v < p - 1 < 1$. Now $q = (K - 1 - v)/X > 0$, so $0 < P_2 < 1$ and $q < 1$. The remaining original, stage, receiver, and core bounds follow exactly as in the preceding construction; in particular $S_{12} = 1 - Vv > 0$ and $S_{22} \ge -1 - Tv \ge -p$.

In both cases first and middle columns are those given by Lemmas Appendix B.6 and Appendix B.9 at the same p, K, E, r. Hence this is a single matrix, not a combination of separately optimized completions. Finally, decreasing the wall coordinate from $m/2$ to $f/2$ preserves every inequality of (B.22): the two oblique coefficients are positive and all other bounds are upper bounds. The two explicit constructions apply again at $r = f/2 > 2$. □

Lemma Appendix B.14 (Exactness at high polygon maxima)**.** *If the maximum of $x_r + y$ on* (B.22) *exceeds H_0, it is attained by one complete canonical source at the fixed η, E. Thus the polygon computes the exact maximum on this high-value range; no assertion about recovery of arbitrary low-value polygon points is made.*

Proof. Take a maximizing point. The first and middle columns are reconstructed at $k = K$ and its actual $r = x_r$. Since $x_r \geq (x_r + y)/2 > 2$, the first and middle columns alone give $k \leq 3 - \beta$ and $x_r \leq 2 + \beta - T$. Thus $x_r \leq \mathcal{R}_{\text{mid}} < 5/2$ and $y > 0$. A wall point was handled in Lemma Appendix B.13. At a nonwall point with $y = C_0$, the second oblique inequality gives $x_r \leq R_T(b_0)$; take $b = b_0, v = 0, q = 1, w = -C_0$. The same entrywise checks apply, and $O_{22} = -1$. This would contradict the positive-receiver lemma at high value, so this alternative cannot actually occur.

If $x_r > y$ and $y < C_0$, the second oblique inequality must be an equality. Otherwise a sufficiently small move in the direction $(-\tau, \Delta)$ preserves the first oblique constraint, improves the bound on x_r, preserves all strict bounds, and increases the objective by $\Delta - \tau > 0$. Set

$$b = (B_c - \delta y)/\ell, \qquad v = (b - b_0)/\delta, \qquad q = 1 - \beta v, \qquad w = -y.$$

The inequalities and the two cofactor identities give $b_0 < b \leq K$, $R_T(b) = x_r$, and $C_T(b) = y$. Thus $P_2 = 1, O_{02} = -1, O_{12} = 1, O_{22} = -1$. The bounds $v \leq p-1$, $q > 0$, and all stage bounds are as in the first wall construction. For the last core entry use $-y + Eb \geq -y \geq -x_r \geq -K$ and $-y + Eb \leq O_{22} = -1 < K$. This proves complete recovery. □

Appendix B.4. Why a maximum at fixed head has a wall representative

The next finite statement is an explicit computational input. Its analytic role is separated from the existence and derivative argument. For $i = X, Y$ set

$$c_X = \frac{\mu}{VX} + \frac{Tt}{V}, \qquad c_Y = \frac{tZ}{Y}, \qquad D_i = E + c_i, \qquad \sigma = Q_o - \ell,$$

and define

$$\begin{aligned} N_{i1} &= (Q_o + \delta E) D_i b - \sigma(D_i a + k), \\ N_{i2} &= Q_o \nu (t + E) D_i b - \sigma[t\nu D_i a + \zeta k]. \end{aligned} \tag{B.23}$$

Proposition Appendix B.15 (Finite exclusion of the nonwall stationary source)**.** *Consider a complete source with $F \geq H_0$, $\Gamma > \nu$, $\sigma > 0$, $r + w > 0$, and the simultaneous contacts*

$$\begin{gathered} O_{10} = 1, \quad O_{20} = -1, \quad D_{21} = k, \quad pZ = 1 + T, \quad O_{11} = 1, \\ P_2 = 1, \quad O_{02} = -1, \quad O_{12} = 1, \quad O_{22} = -1, \\ (S_{00} = p \text{ or } S_{10} = p), \qquad (O_{01} = -1 \text{ or } O_{21} = 1). \end{gathered} \tag{B.24}$$

Then all four numbers $N_{X1}, N_{Y1}, N_{X2}, N_{Y2}$ are positive.

Certificate reduction and verification obligation. The identity

$$N_{i2} = \zeta N_{i1} + E\Gamma\sigma D_i(b - a)$$

reduces the assertion to $N_{X1} > 0$ and $N_{Y1} > 0$. Here $\zeta = t\ell + EQ_o > 0$, and $b = 1 + X + \delta v > 1 + X$, whereas $a < p - X < 2 - X$; thus $b - a > 0$. The two choices in each of the last line of (B.24) give four contact systems. Each is combined separately with $-N_{X1} \geq 0$ and $-N_{Y1} \geq 0$, giving eight closed necessary systems.

Their algebraic elimination is explicit. The first column is

$$\begin{aligned} S_{00} = p: \quad & k = \frac{p\mu + X(V + T)}{\mu + X(VE + Tt)}, \quad j = (k - p)/X, \\ S_{10} = p: \quad & k = \frac{p\mu + Y(V + T)}{V(tZ + EY)}, \quad j = (tk - p)/Y, \end{aligned}$$

with $g = (tk - 1 - Yj)/V$. The middle column is respectively

$$\begin{aligned} O_{01} = -1: \quad & r = \frac{Q_o k + E[\delta(1+Y) + \Gamma(1-X)]}{Q_o + \delta E}, \\ & h = (r - 1 - Y - t(1-X))/Q_o, \quad a = 1 - X + \delta h; \\ O_{21} = 1: \quad & h = (k - 1 - Z)/\nu, \quad r = k - Ea, \\ & a = (k - 1 - Y + \Gamma h)/(t + E). \end{aligned}$$

In either case

$$b = \frac{\delta(r - 1 + Y) - \Gamma(1 + X)}{Q_o}, \quad v = (b - 1 - X)/\delta,$$
$$q = 1 - \beta v, \qquad y = 1 + Z + Eb - \nu v.$$

Every denominator displayed is positive. The archived finite systems retain these as denominator-cleared equations and retain defined auxiliaries, so the maximum input degree is three; no auxiliary is free to vary independently of its defining equation.

For precision, their common closed root box is

t	$[1/16, 1]$	E	$[1/512, 1/2]$
β, T	$[1/16, 7/8]$	V	$[1/16, 1]$
X, Z	$[1/2, 1]$	Y	$[1/32, 1]$
δ	$[1/8, 1]$	Γ, ν	$[1/16, 1]$
$Q_o + \delta E$	$[0, 2]$	k	$[33/16, 3]$
a	$[1/8, 3/2]$	b	$[3/2, 3]$
r	$[33/16, 5/2]$	y	$[13/8, 5/2]$
p	$[17/16, 15/8]$	g	$[1/8, 15/16]$
j	$[3/16, 1]$	h	$[1/16, 1]$
v	$[0, 1]$	D_i	$[0, 48]$.

The less immediate bounds follow without numerical root searching. Put $\epsilon = r - 2 \geq 1/16$. The middle column gives $\nu h \geq \epsilon$, hence $\nu \geq 1/16$; $\Gamma > \nu$, $V > \Gamma$, and $Y > XV$ give the displayed lower bounds for Γ, V, Y. The strict first-column gap gives

$$g - 1/8 = (k - 33/16) + (1 - Xj) + (15/16 - O_{00}) > 0.$$

Also $p < 15/8$ and $k - Xj \leq p$ imply $j > 3/16$. The contact $O_{20} = -1$ gives $Ek - \nu g = (1 - Z) + Z(1 - j - \beta g) \geq 0$, so $E > 1/384 > 1/512$. Further,

$$(t - E)a - 2\epsilon = (1 - Y) + (1 - Z) + (\Gamma - \nu)h + (1 - O_{21}) > 0,$$

giving $a > 1/8$, and $T(1-h) - \epsilon - Ea = (1 - O_{21}) + Z(p - 1 - \beta h) \geq 0$ gives $T > 1/16$. Finally $k \leq 3 - \beta$ and $r \leq 2 + \beta - T$ imply $r < 5/2$, whence $y = F - r > 13/8$. The definitions bound $D_X < 1/2 + 32 + 14 < 48$ and $D_Y < 1/2 + 32 < 48$. Other endpoints are weakened forms of the common-column and first-column height bounds.

The eight proof trees, grouped by stage contact, middle contact, and tested numerator X, Y, have respective node counts

$$(17317, 9111), \quad (42915, 22627), \quad (8781, 4825), \quad (20223, 10711).$$

They contain 136510 nodes and 68259 contradictory leaves, with no open leaf. Each stored model has 18 equalities and 60 nonnegative polynomial rows, including the one tested negative numerator. Their coefficients, root boxes, and immutable payload identities are specified in the supplementary source index under the nonwall stationary exclusion (historical identifier C19/V20). Acceptance reconstructs those rows and checks every leaf with outward integer interval arithmetic; covered binary splits propagate emptiness to each root by finite induction. This finite acceptance, rather than the preceding algebra alone, establishes the two strict positivities. The displayed numerator identity then establishes the other two. □

Theorem Appendix B.16 (Same-head maximum on the wall)**.** *If the complete canonical fibre at a fixed head has a source of height greater than H_0, its maximum is attained, and every value of E attaining that maximum has a complete wall representative of the same height. Consequently every attainable target $H_0 < f < 5$ has a same-head wall representative of height exactly f.*

Proof. Attainment follows directly from the full physical domain. Restrict to heights $F \geq f > H_0$ below a known attainable height. Then $r \geq f/2$, $k \geq r$, $p \geq k/2 \geq f/4$, and $p \leq 2, k \leq 4$; all other inner variables are bounded by the physical bands. At a fixed head the six positive cofactors are constant. The remaining constraints are closed, and the pivots have uniform positive lower bounds. This high-value fibre is compact.

Normalize p, k and use the exact polygon maximum. Write $\sigma = Q_o - \ell$ and $R_m = \mathcal{R}_{\mathrm{mid}}$. For $\sigma \geq 0$, elimination of $y = F - x_r$ from the polygon gives its high maximum as the minimum of

$$\begin{gathered} U_1 = 2R_m, \quad U_2 = 2\mathcal{C}/(\Delta + \tau), \quad U_3 = R_m + C_0, \\ U_4 = [\mathcal{C} + (\Delta - \tau)C_0]/\Delta, \\ U_5 = (\mathcal{D} + \sigma R_m)/Q_o, \qquad U_6 = a_T + c_T + \sigma K/\delta. \end{gathered}$$

For example, set $s_0 = R_m - x_r$, $s_1 = x_r - y$, $s_2 = C_0 - y$, $s_3 = \mathcal{C} - \Delta x_r - \tau y$, and $s_4 = \mathcal{D} - \ell x_r - Q_o y$. Their slack identities are

$$\begin{gathered} U_1 - F = 2s_0 + s_1, \quad U_2 - F = [2s_3 + (\Delta - \tau)s_1]/(\Delta + \tau), \\ U_3 - F = s_0 + s_2, \quad U_4 - F = [s_3 + (\Delta - \tau)s_2]/\Delta, \\ U_5 - F = (s_4 + \sigma s_0)/Q_o, \quad U_6 - F = [(\Delta - \tau)s_4 + \sigma s_3]/(\Omega\delta). \end{gathered}$$

The converse follows by comparing the lower and upper bounds for x_r, so these are complete polygon budgets, not a selection of numerically observed active constraints.

Suppose there is no wall maximum at a maximizing E_*. If $\sigma \leq 0$, the direction $(-1, 1)$ preserves height, improves both oblique bounds and the upper bound on x_r, and reaches either the wall or $y = C_0$. The latter has a complete high-value recovery with $v = 0$, already excluded. Hence $\sigma > 0$. The first four budget equalities force, respectively, a wall point or $y = C_0$. They therefore have strict slack at this maximum. Only U_5, U_6 can be active.

We next verify the increase of U_6 where needed. By the strict first-column gap, the only locally relevant capacities are the four in (B.28); two are constant in E, and the others have the form $n/(E + c_i)$, $i = X, Y$. For these last two branches

$$\frac{d}{dE}\left(a_T + c_T + \frac{\sigma n}{\delta(E + c_i)}\right) = \frac{n[\delta c_i - (Q_o - \nu)]}{\delta(E + c_i)^2} > 0.$$

Here is a direct proof of the sign. The inequalities $X > \delta$, $\nu + Z > 1$ imply $\Delta > X\delta$: for $Z \geq X$ use $\Delta - X\delta = \delta(Z - X) + X\nu$; for $Z \leq X$ use

$$\Delta - X\delta = X(\nu + Z - 1) + X(1 - X) + (X - \delta)(X - Z).$$

Thus $\delta Z/X - (Q_o - \nu) = \Delta/X - Q_o > 0$. If $d = tX - Y \geq 0$, both $c_X, c_Y \geq Z/X$. If $d \leq 0$, then $c_X \geq c_Y$, and positivity for c_Y follows from $\Gamma + \nu + t\delta(Z/Y - 1) > 0$ when $Z \geq Y$, or from

$$\Gamma + (\nu + Z - 1) + (1 - Y) + (1 - t\delta/Y)(Y - Z) > 0$$

when $Z \leq Y$; here $t\delta < Y$. Constant capacities likewise give a positive derivative. All finitely many active right branches of U_6 increase.

It follows that U_5 must be active. Its slack identity gives $x_r = R_m$ and $\ell x_r + Q_o y = \mathcal{D}$. The nonwall construction in Lemma Appendix B.14 therefore gives all four tail contacts in (B.24). The common middle reconstruction gives $O_{11} = 1$, with $O_{01} = -1$ or $O_{21} = 1$.

On an active differentiable branch of R_m,

$$U_5' = b + \frac{\sigma}{Q_o} R_m', \qquad b = \delta(R_m - a_T)/Q_o > 1 + X > a, \qquad 0 < \sigma/Q_o < 1.$$

For the five middle capacities independent of K, the derivatives are zero or, at their actual contacts,

$$-\frac{ta}{t+E}, \qquad -\frac{(\Gamma + t\beta X)a}{\nu + \Gamma + t\beta X + \beta X E}.$$

Each is at least $-a$, so it cannot prevent U_5 from increasing. For a constant active K, the two core intersection derivatives are likewise $-Q_o a/(Q_o + \delta E)$ and $-ta/(t+E)$. Thus a blocking right branch must use an E-dependent first capacity and an active middle core capacity. These force $O_{20} = -1$, $D_{21} = k$. The strict first-column gap excludes $O_{00} = 1$ and $g = 1$; hence they also force $O_{10} = 1$ and $S_{00} = p$ or $S_{10} = p$.

For completeness the sign region of this source is determined by three identities. Put $G = g - v$, $A = k - b$, $s_P = 1 + P_0$. The contacts just obtained give

$$\Gamma G = (r-2) + tA + Y s_P > 0, \qquad (t-E)k = (Y+Z)j + (V+T)g > 0,$$

$$(\Gamma - \nu)G = (r+w) + (t-E)A + (Y+Z)s_P > 0.$$

Thus $\Gamma > \nu$. In particular the region $\Gamma \le \nu$, including equality, requires no finite nonwall exclusion.

The two possible variable-K core derivatives are

$$R_{12}' = \frac{Q_o(K_i' - a)}{Q_o + \delta E}, \qquad R_{22}' = \frac{\zeta K_i'}{\nu(t+E)} - \frac{ta}{t+E}, \qquad K_i' = -\frac{k}{D_i}.$$

Multiplying U_5' by the respective positive denominators gives exactly N_{i1} and N_{i2} in (B.23). At a local maximum at least one common active right branch must have a nonpositive numerator; otherwise the finite minimum of all active budgets increases. Proposition Appendix B.15 excludes this.

No differentiability assumption at a capacity tie is needed: take all active right branches of the finite family of rational functions. There is a right neighbourhood since a high source has $E < 1/2$ and $\ell > 0$. At $E = 0$, the bare middle capacity C_2 ties the two core intersection capacities at K. Its right derivative is $K_i' - a_0$, where $a_0 = (r - 1 - Y)/t$; the actual reconstructed $a = a_0 + \Gamma h/t > a_0$ supplies an intersection branch with no greater derivative. Thus it introduces no missing obstruction. The hypothetical absence of a wall representative is impossible. Finally lower a maximum wall to any prescribed $f > H_0$ by Lemma Appendix B.13. □

Appendix B.5. The head entrance used by the near-peak argument

Proposition Appendix B.17 (Finite nonpositive-difference exclusion)**.** *A complete canonical source with $d = tX - Y \le 0$ has $F \le H_0$. This includes $d = 0$.*

Analytic reduction to eight finite systems. Suppose $F > H_0$. Theorem Appendix B.16 supplies a same-head maximum wall of height $m > H_0$. At a normalized wall feasibility is exactly $r \le \mathcal{R}_{\mathrm{mid}}(E), \mathcal{W}(E)$. The first capacity is nonincreasing in its high range. When $d \le 0$, the second is nondecreasing: C_0 increases, and

$$\left(\frac{\mathcal{D}}{\ell + Q_o}\right)' = \frac{\delta Q_o(c_T - a_T)}{(\ell + Q_o)^2} > 0.$$

For the complete first-column terms $N/(A + BE)$, differentiation of ΩK_i gives a positive multiple of $\tau/X - (t - V) = -d/X$, or $\tau t/Y - (t - V) = -Vd/Y$, when $B > 0$; when $B = 0$ the derivative is positive. Hence their minimum is nondecreasing.

These directions allow a maximum wall with $\mathcal{R}_{\mathrm{mid}}(E) = \mathcal{W}(E) = m/2$. Indeed $\mathcal{W}(0) \le 1 + Z \le 2 < m/2$, whereas at an actual maximum both capacities are at least $m/2$. Continue the two continuous capacities to $E_{\max} = \min\{1, \nu/\delta\}$. At the right endpoint $\mathcal{R}_{\mathrm{mid}}$ cannot remain above $H_0/2$ while $\mathcal{W} \ge m/2$: near-endpoint wall recovery and a convergent subsequence of bounded inner variables would give a complete high wall with either $\ell = 0$ or $E = 1$, contrary to the proved high-value bounds. The intermediate value theorem gives an interior intersection at height at least $m/2$; maximality gives equality.

At this intersection the C_0 capacity cannot be active, by the strict last slack of Lemma Appendix B.12. The two other tail-capacity slack identities force

$$O_{02} = -1, \quad O_{12} = 1, \quad O_{22} = -1, \qquad P_2 = 1 \text{ or } b = k.$$

Moreover

$$\frac{k-p}{X} - \frac{tk-p}{Y} = \frac{(Y - tX)(k-p) + pX(1-t)}{XY} \ge 0.$$

The common first-column completion therefore gives $S_{00} = p$, $O_{10} = 1$, and $P_0 = -1$ or $O_{20} = -1$. The middle completion gives $O_{11} = 1$ and $O_{01} = -1$ or $O_{21} = 1$. Together with $pZ = 1 + T, w = -r, r \ge 33/16, d \le 0$, these are eight exhaustive contact systems. Ties belong to multiple systems and are not discarded.

The finite certificates for these eight systems have, in the order $(P_0 = -1, O_{20} = -1)$, then $(O_{01} = -1, O_{21} = 1)$, then $(P_2 = 1, b = k)$, node counts

$$21341, 31141, 101533, 313597, 21593, 30615, 106077, 284869.$$

The total is 910766 nodes, 455387 contradictory leaves, and zero open leaves. The exact source systems and payload identities are indexed as the nonpositive head-difference exclusion (historical identifier D22/V22). For the $P_2 = 1$ systems there are 24 equalities and 50 nonnegative rows; for the $b = k$ systems there are 23 equalities and 49 nonnegative rows. All input polynomials have degree at most three. These finite acceptance results are computational premises of this proposition.

We record the source of their compact root boxes, to distinguish coverage from a box chosen around numerical samples. Write $\epsilon = r - 2 \ge 1/16$. The contacts give

$$(t - V)v = \epsilon + (1 - t) + (Y - tX)q, \qquad tp = 1 + Vg + (Y - tX)j,$$

$$\mu(g - v) = (Y + Z)\epsilon + (tZ + YE)(k - b) + Y(1 + O_{20}) + Y(1 + O_{22}),$$
$$\mu(1 - p + g) = (\Delta + \tau)\epsilon + \Omega(k - b) + \tau(1 + O_{20}) + \tau(1 + O_{22}).$$

The common first-column budget gives $k \le 9/4$ and $16X^2 - 17X + 4 \ge 0$. This polynomial is negative on $[1/2, 45/64]$, so $X > 45/64$. Since $v \le p - 1 < 7/8$, the first identity gives $t > 17/30 > 9/16$, $t - V > 1/14$, and $v > \epsilon$. The middle column gives $\nu \ge 1/16$, $T > 1/16$, then $\beta > 1/8$, $Z > 1/2$. With $Y \ge tX$, this yields $Y > 25/64$, $V > 3/64$, $\tau > 3/64$, $\mu = \Gamma Z + \nu Y > 3/128$, and $\Delta > \tau + 1/32 > 5/64$. The strict first-column gap gives $g > 1/8, j > 3/16$. On the wall

$$Eb - \epsilon = (1 + O_{22}) + (1 - Z) + Z(1 - P_2) + \nu v > 0,$$

so $E > 1/36 > 1/64$. The other upper bounds are the physical boxes, $a < p - X < 3/2$, $b \le k \le 9/4$, and the first-column height bounds of Lemma Appendix B.2, equation (B.7),

and Lemma Appendix B.7. The archive uses the following closed weakenings:

t	$[9/16, 1]$	E	$[1/64, 1/2]$
β	$[1/8, 7/8]$	V	$[3/64, 1]$
T	$[1/16, 7/8]$	X	$[45/64, 1]$
Y	$[25/64, 1]$	Z	$[1/2, 1]$
δ	$[1/8, 1]$	Γ	$[0, 1]$
ν	$[1/16, 1]$	τ	$[3/64, 1]$
Δ	$[5/64, 1]$	μ	$[3/128, 1]$
k, r	$[33/16, 9/4]$	a	$[0, 3/2]$
b	$[3/2, 9/4]$	p	$[17/16, 15/8]$
g	$[1/8, 15/16]$	j	$[3/16, 1]$
h	$[1/16, 1]$	v	$[1/16, 7/8]$
q	$[1/16, 1]$.		

The stated lower bound for q uses the high-value positive-tail bound in (B.7) of Lemma Appendix B.2, namely $q > 41/480 > 1/16$; this is one of the root-entry dependencies and is not inferred merely from $q > 0$. All six cofactor auxiliaries remain bound to their definitions. As for Proposition Appendix B.15, full integer interval acceptance and covered split induction exclude each closed necessary system. Hence a complete source with $d \le 0, F > H_0$ cannot exist. □

Corollary Appendix B.18 (The two remaining head entrance inequalities)**.** *At a head which admits height $f > H_0$, put $p = (1 + T)/Z$, $d = tX - Y$. Then*

$$d > 0, \qquad tp + d - 1 > 0, \qquad 1 + t + d < 33/16.$$

Proof. The first statement follows from Proposition Appendix B.17. At a normalized complete representative $O_{10} = 1$, and

$$tp + d - 1 = (O_{10} - 1) + t(p - S_{00}) + Vg + d(1 - j) > 0.$$

Let $r_0 = 1 + t + d$ and take the same-head wall of height f, with $r = f/2$. If $r \le r_0$, replace only its tail by $b = b_0 = 1 + X, v = 0, q = 1, w = -r$. The resulting original entries are

$$O_{02} = -1, \qquad O_{12} = 1 + r - r_0, \qquad O_{22} = C_0 - r - 1.$$

Here $O_{12} > 0$ because $r > 2$, $t \le 1$, and $0 < d < 1$, while $O_{12} \le 1$ by the supposed bound on r. Also $C_0 > r$ by the strict wall slack, and $Eb_0 \le EK < 1 - T$, giving $-1 < O_{22} < 0$. As $b_0 \le 2 < K$, all remaining stage, receiver, prefix, and core bounds follow as in the wall construction. This would be a complete source of height $f > H_0$ with $v = 0$, impossible. Thus $r > r_0$.

If nevertheless $r_0 \ge 33/16$, keep the same E, p, K and first column, reconstruct the middle column at $r' = r_0$, and use the same zero-receiver tail with $w = -r_0$. The middle reconstruction is valid because $2 < r_0 < r \le \mathcal{R}_{\mathrm{mid}}$; the tail check now uses $C_0 > r > r_0$. It gives a complete source of height $2r_0 \ge H_0$ and $v = 0$, again impossible. Therefore $r_0 < 33/16$. □

Appendix B.6. From a wall representative to a single target endpoint

We now isolate the last finite input to the representative theorem. It concerns a specified set of complete matrices, not all matrices having one of the listed contacts. Let $\mathcal{R}_*$ be the subset of $\mathcal{K}$ satisfying

$$\begin{gathered} w = -r, \quad pZ = 1 + T, \quad O_{10} = O_{11} = 1, \\ (O_{02}, O_{12}, O_{22}) = (-1, 1, -1), \\ (S_{00} = p \text{ or } S_{10} = p), \quad (P_0 = -1 \text{ or } O_{20} = -1), \\ (O_{01} = -1 \text{ or } O_{21} = 1), \quad (P_2 = 1 \text{ or } b = k). \end{gathered} \tag{B.25}$$

Every contact here is evaluated in the same matrix (B.1).

Proposition Appendix B.19 (Finite exceptional-contact exclusion)**.** *For a complete source in $\mathcal{R}_*$,*

$$S_{00} = p \text{ or } P_0 = -1 \quad \Longrightarrow \quad F < \Lambda = 1653/400.$$

This is a computational proposition, established by the six exact exclusion certificates described below.

Source reduction and certificate acceptance. Suppose $F \geq \Lambda$. The analytic entrance results above apply since $\Lambda > 33/8$. The possible exceptional contacts are covered by the following six systems; no disjointness is required at ties.

System	First-column contact	Stage contact	Middle contact
PX1	$P_0 = -1$	$S_{00} = p$	$O_{01} = -1$
PX2	$P_0 = -1$	$S_{00} = p$	$O_{21} = 1$
PY1	$P_0 = -1$	$S_{10} = p$	$O_{01} = -1$
PY2	$P_0 = -1$	$S_{10} = p$	$O_{21} = 1$
EX1	$O_{20} = -1$	$S_{00} = p$	$O_{01} = -1$
EX2	$O_{20} = -1$	$S_{00} = p$	$O_{21} = 1$

In each system retain all complete physical inequalities and all common contacts in (B.25). The last disjunction is encoded by $(1 - P_2)(k - b) = 0$; both factors are nonnegative. The trigger is the exact integer inequality $800r - 1653 \geq 0$. Thus it is not rounded to the earlier threshold 33/8.

For explicit source-to-box inclusion put $\epsilon = r - 2$, $A = \delta h$, $B = \epsilon + Ea$, and $u_1 = 1 + O_{01}$. These three additional coordinates and the three cofactors below are bound to their defining polynomials. The frozen 24-coordinate root boxes are

Coordinate	Interval	Coordinate	Interval
t	$[15/32, 1]$	E	$[1/64, 1/2]$
β	$[1/8, 7/8]$	V	$[0, 1]$
T	$[1/16, 7/8]$	X	$[45/64, 1]$
Y	$[0, 1]$	Z	$[1/2, 1]$
δ	$[1/8, 1]$	Γ	$[0, 1]$
ν	$[1/16, 1]$	k	$[33/16, 9/4]$
a	$[0, 3/2]$	b	$[1, 9/4]$
r	$[33/16, 9/4]$	p	$[17/16, 15/8]$
g	$[1/8, 15/16]$	j	$[3/16, 1]$
h	$[1/16, 1]$	v	$[0, 7/8]$
q	$[1/16, 1]$	A	$[0, 1]$
B	$[1/16, 1/4]$	u_1	$[0, 1]$

Here $\epsilon > 1/16$, $r \leq k \leq 9/4$, and $\beta < 5 - F < 7/8$. The normalized middle-column bands imply

$$\nu h \geq \epsilon + Ea, \qquad \beta h \leq s,$$
$$\epsilon + Ea \leq \nu h - (1 - Z) \leq \nu s/\beta - (1 - Z) = T(\beta - s)/\beta < T \leq s = p - 1.$$

Consequently $\nu, T, h > 1/16$, $\beta > 1/8$, $17/16 < p < 15/8$, and $Z = (1 + T)/p > 1/2$. Also $\delta > 1/8$, $g < 3 - r < 15/16$, and $k \leq p + Xj$ gives $j > 3/16$. The first-column bound $O_{00} < 3 - r$, together with $Xj < 1$, gives $g = k - Xj - O_{00} > 1/8$.

To obtain the stated lower bound for X, use $X > 1/(1 + \beta) > 1/2$ and the common first-column budget $k \leq 1 + \beta + X(1 - \beta^2)$. The identity

$$X(1 + X - k) + \tfrac{1}{4} = X[1 + \beta + X(1 - \beta^2) - k] + (X\beta - \tfrac{1}{2})^2 \geq 0$$

gives $16X^2-17X+4\ge 0$. This convex quadratic is negative at both endpoints of $[1/2,45/64]$, and hence throughout that interval. Thus $X>45/64$. The contact $O_{02}=-1$ gives $b=1+Xq+v>1$ and $v\le p-1<7/8$. The contact $O_{12}=1$ gives $r<1+(9/4)t$, so $t>17/36>15/32$. The positive wall slack gives $Eb>\epsilon$, hence $E>1/36>1/64$; the common budget $Ek<1-T$ gives $E<1/2$. The middle stage gives $a<p-X<3/2$, and $r+Ea\le k$ gives $1/16<B\le 1/4$. Since $X>\delta$ and $h\le 1$, $O_{01}=-a-X+\delta h<0$; therefore $0\le u_1<1$. Finally $q>41/480>1/16$, while $P_2\le 1$ gives $q<1$. The remaining bounds follow directly from (B.2). Closed endpoints in the model enlarge this necessary system and do not remove any strict physical source.

For completeness, the additional polynomial inequalities used in the six models have explicit same-source payments. Put $L=1+B-X>0$. The normalized bottom prefix and the middle bands give

$$Z\{h(\beta-s)-B\}=(1-O_{21})+(1-Z)(1-h+B)\ge 0,\quad h\le s+u_1,\quad \delta s\ge L.$$

Define

$$\begin{aligned}F_1&=(1+Xu_1)^2-4XA-4X^2B,\\ F_2&=A-4XAL-4X^2BL,\\ F_3&=(1+Xu_1)^2(1-4XL)-4X^2B.\end{aligned}$$

Their nonnegativity follows from

$$\begin{aligned}F_1&=(2Xh-1-Xu_1)^2+4X^2[h(\beta-s)-B]+4X^2h(s+u_1-h),\\ AF_2&=[2\delta(A+XB)-A]^2+4XA(A+XB)(\delta s-L)\\ &\quad+4X\delta^2(A+XB)[h(\beta-s)-B],\\ F_3&=4XF_2+(1-4XL)F_1.\end{aligned}$$

Indeed $A>0$, and $F_2\ge 0$, $B,L>0$ imply $1-4XL>0$. These identities use $A=\delta h$ and $u_1=1-a-X+A$. If $O_{01}=-1$, so $u_1=0$, $F_3\ge 0$, $B>1/16$, and $X>1/2$ further imply $X\ge 4/5$, because

$$[4X(1+B-X)+4X^2B-1]_{B=1/16}=-\tfrac{1}{4}(3X-1)(5X-4).$$

This last strengthening is used only in the three systems with $O_{01}=-1$. All equalities and inequalities remain polynomial conditions on one 24-coordinate point.

Two further strict rows in the model also come from this same source:

$$\begin{aligned}\mu(g-v)&=(Y+Z)\epsilon+(tZ+YE)(k-b)+Y(1+O_{20})>0,\\ (t-E)k&=O_{10}+O_{20}+(Y+Z)j+(V+T)g>0.\end{aligned}$$

The first uses the three tail contacts and $O_{10}=1$; the second uses $O_{10}=1$, $O_{20}\ge -1$, and positive receivers. Thus $g>v$ and $t>E$ are necessary rows, not extra assumptions.

The model files specify the complete coefficient lists and their necessary-row derivations. The exact acceptor rebuilds interval bounds and polynomial derivative enclosures. An interval leaf exhibits an empty necessary interval. At a combination leaf, after normalizing the current box to $z_i\in[-1,1]$, it reconstructs necessary rows $b_j+\sum_i a_{ji}z_i\ge 0$ and verifies rational weights $\lambda_j\ge 0$ with

$$\sum_j \lambda_j b_j+\sum_i\left|\sum_j \lambda_j a_{ji}\right|<0.$$

The left side bounds the weighted sum above on the entire box, contradicting the necessary rows. Each split covers its closed parent; induction through every branch therefore proves root

emptiness. The six accepted trees contain respectively $28073, 38039, 42425, 49053, 29989, 40963$ nodes, and together 114274 contradiction leaves with no open leaf. Their exact files and replay entry are identified in Supplementary Index S3. The argument above supplies the source inclusion and the logical meaning of acceptance; the finite verification of all those leaves is a separate part of the computational proof. □

Lemma Appendix B.20 (Strict selection of the first-column capacity)**.** *Every source in* $\mathcal{R}_*$ *with* $F \geq \Lambda$ *satisfies*

$$S_{10} = p, \quad O_{20} = -1, \quad S_{00} < p, \quad P_0 > -1, \qquad k = K_{24} < K_{13}, K_{14}, K_{23},$$

where the capacities are those in (B.28)*.*

Proof. Proposition Appendix B.19 and the disjunctions in (B.25) give the four entry conditions. Write $K_i = N_i/D_i$, with each $D_i > 0$. Direct expansion in the common matrix gives

$$\begin{aligned}
N_{13} - D_{13}k &= \Gamma(p - S_{00}) + XV(1 + P_0) + \beta X(1 - O_{10}) > 0,\\
N_{14} - D_{14}k &= \mu(p - S_{00}) + XV(1 + O_{20}) + XT(1 - O_{10}) > 0,\\
N_{23} - D_{23}k &= \Gamma(p - S_{10}) + VY(1 + P_0) + \beta Y(1 - O_{10}) > 0,\\
N_{24} - D_{24}k &= \mu(p - S_{10}) + VY(1 + O_{20}) + YT(1 - O_{10}) = 0.
\end{aligned}$$

All strict signs follow from the positive cofactors and positive X, Y, V. Dividing by the positive denominators proves the claim. □

Appendix B.7. The prescribed-target representative and its equivalence

The preceding analytic lemmas and the explicitly identified finite exclusions give the following representative theorem. It asserts existence at the same head, not that every original source already has the representative contacts.

Proposition Appendix B.21 (Prescribed-target representative)**.** *Put* $\Lambda = 1653/400$*. Fix a seven-coordinate head* $\eta = (t, \beta, V, T, X, Y, Z)$ *for which a complete source in* $\mathcal{K}$ *can have height at least* f*, where* $\Lambda \leq f < 5$*. There is a complete matrix at this same head, with height exactly* f*, satisfying*

$$\begin{gathered}
w = -r, \quad r = f/2, \quad pZ = 1 + T, \quad P_1 = -1, \quad P_2 = 1,\\
O_{10} = O_{11} = 1, \quad (O_{02}, O_{12}, O_{22}) = (-1, 1, -1),\\
S_{10} = p, \quad O_{20} = -1, \quad S_{00} < p, \quad P_0 > -1.
\end{gathered} \tag{B.26}$$

Its cofactor and physical inequalities remain those of Definition Appendix B.1. It also satisfies the high-value qualifications recorded in (B.31) *below. Conversely, a target representative at this head is a complete physical source of height* f*.*

Proof. Fix the head η and a target $\Lambda \leq f < 5$, and assume that some complete source has $F \geq f$. Compactness is applied before imposing the strict endpoint qualifications. On the high-value part of the fixed-head fibre,

$$r \geq F/2 \geq f/2, \quad k \geq r + Ea \geq r, \quad p \leq 1 + \beta \leq 2, \quad k \leq p(1 + X) \leq 4.$$

Thus $p \geq f/4 > 1$, $k, r > 2$; all remaining inner coordinates are bounded by the original and stage bands. The six positive cofactors are fixed with the head. The polynomial weak physical inequalities and the uniform positive pivot bounds make this fibre compact. Height therefore attains a maximum $m \geq f$.

Theorem Appendix B.16 supplies a complete wall representative at the same head and height m. The wall polygon and reconstruction in Lemma Appendix B.13 then give height

exactly f: lowering its two equal wall coordinates from $m/2$ to $f/2$ preserves every positive-coefficient upper inequality of the polygon. This use of reconstruction is confined to the high range $f/2 > 2$. The set of such complete walls of height f is a nonempty closed subset of the compact fibre. Choose one minimizing E, and write $r = f/2$.

Normalize its common first and middle columns by Lemmas Appendix B.5, Appendix B.6, and Appendix B.9, without changing the head, E, r, w. The resulting first-column capacity is the full $K(E)$. The high-value entrance inequalities give $v, \ell > 0$, $Ek < 1 - T$, and $Eb > r - 2$; hence $0 < E < 1/2$, strictly within the admissible E-interval. In the notation of Lemma Appendix B.13, feasibility requires $r \le \mathcal{R}_{\mathrm{mid}}(E), \mathcal{W}(E)$. If $\mathcal{W}(E) > r$, continuity preserves that strict inequality when E decreases a little. Lemma Appendix B.11 preserves first- and middle-column feasibility. The full wall reconstruction would therefore yield the same height at smaller E, contradicting minimality. Thus $\mathcal{W}(E) = r$.

On this wall $y = r$, and the last identity in (B.21) gives $C_0 - y > 0$, since $\ell v > 0$ and all other terms are nonnegative. Thus $C_0 > r$, so the active tail bound is not C_0. If the active bound is $\mathcal{C}/(\Delta + \tau)$, its slack $J_a + \Omega(K - b) = 0$ forces $J_a = 0$ and $b = K$. If it is $\mathcal{D}/(\ell + Q_o)$, its slack $\ell J_a + \Omega J_b = 0$ forces $J_a = J_b = 0$. All the coefficients in these sums are positive. It follows that

$$(O_{02}, O_{12}, O_{22}) = (-1, 1, -1), \qquad P_2 = 1 \text{ or } b = k.$$

The common column reconstructions also give the first- and middle-column alternatives in (B.25). Consequently this one complete representative belongs to $\mathcal{R}_*$. Proposition Appendix B.19 and Lemma Appendix B.20 give

$$S_{00} < p, \quad P_0 > -1, \quad S_{10} = p, \quad O_{20} = -1, \qquad k = K_{24} < K_{13}, K_{14}, K_{23}.$$

Suppose now that $P_2 < 1$. Then $b = k$, $J_a = 0$, and $J_b = \Delta(1 - P_2) > 0$. The only active tail bound is $\mathcal{C}/(\Delta + \tau)$; the other two have strict slack. Lemma Appendix B.7 excludes the other eight original capacities. Together with the three strict comparisons above, this makes the full $K(E)$ locally equal to $K_{24}(E)$. Its numerator $N_{24} = p\mu + Y(V + T) > 0$ is independent of E, and direct differentiation gives

$$\frac{d(\Omega K_{24})}{dE} = -\frac{N_{24} V \mu (tX - Y)}{[V(tZ + EY)]^2} < 0.$$

Decreasing E therefore improves the active tail capacity; continuity preserves the other tail and first-column gaps, and Lemma Appendix B.11 preserves the middle-column test. Reconstruction again contradicts minimality. Thus $P_2 = 1$.

The equalities $J_a = J_b = 0$ now give

$$\ell(r - a_T) + Q_o(r - c_T) = 0, \qquad r = a_T + Q_o b/\delta.$$

Since $a_T < 2 < r$, solving them is legitimate and yields $E = L_D$ and $b = \delta(r - a_T)/Q_o$ in (B.29). The strict C_0 slack gives $E > L_C$. The tail receiver is positive, so $b > 1 + X$; the complete core retains $b \le k$. This proves all contacts and qualifications claimed for the prescribed-target representative.

Here is also the reduction to the five simultaneous middle tests, including its converse. Write the two increasing lines as $I_i(h) = A_i + B_i h$, the decreasing lines as $J_i(h) = C_i - D_i h$, and put $R_{ij} = (B_i C_j + D_j A_i)/(B_i + D_j)$. The complete middle test is the minimum of

$$I_1(H),\ I_2(H),\ C_1,\ C_2,\ R_{11},\ R_{21},\ R_{12},\ R_{22}.$$

The common normalization has $H = s/\beta$, and

$$R_{11} = A_1 + B_1 s, \qquad I_1(H) - R_{11} = Q_o s(1 - \beta)/\beta > 0,$$
$$C_1 - R_{11} = (t - Q_o)s > 0.$$

Also $A_2 \le 2 < r$ and

$$B_2(C_2 - R_{22}) = D_2(R_{22} - A_2).$$

Thus the five retained tests in (B.30) imply the three omitted ones, and the reverse implication is immediate. All denominators are positive; at ties the same closed inequalities remain valid. If the minimum capacity is written before selecting K_{24}, the two relevant intersections increase with K:

$$\partial_K R_{12} = \frac{Q_o}{Q_o + \delta E} > 0, \qquad \partial_K R_{22} = \frac{t\nu - E\Gamma}{\nu(t+E)} > 0.$$

Hence the strict capacity qualifications retain every original first-column comparison rather than discarding it.

Conversely, a head satisfying the stated entrance qualifications, (B.29), and (B.30) supplies the formal tail $v = (b - 1 - X)/\delta > 0$, $q = 1 - \beta v$, $w = -r$. The formulas for E, b give all three original tail contacts by substitution. In particular

$$Eb - (r-2) = (1 - Z) + \nu v > 0,$$

so $E > 0$ before any assertion of complete physicality. First apply the five-to-eight line-interval argument purely algebraically and choose its smallest admissible h, then $a = (r - 1 - Y + \Gamma h)/t > 0$. The fourth line inequality gives $r + Ea \le k$, hence $r < k$. The formal tail contacts also give $J_{\text{tail}} = J_a = 0$. These facts, $b \le k$, and the four capacity qualifications are exactly the hypotheses of Lemma Appendix B.8; they restore the full twelve-capacity first-column condition without assuming a pre-existing complete matrix. Lemma Appendix B.9 now supplies the complete middle column, and Lemma Appendix B.13 supplies the tail. Their common realization is exactly the one printed below (B.30), with every initial, stage and core band satisfied. Its height is $2r = f$. No step selects incompatible completions of different columns, assumes that all maxima share these contacts, or uses the final global height bound. □

For clarity, the exact endpoint qualifications used in this proposition are included here. They can either be retained in these coordinates or replaced by the equivalent scale predicate in Proposition Appendix B.23. For the fixed head put

$$p = (1+T)/Z, \quad s = p - 1, \quad Q_o = t\delta - \Gamma, \quad \ell = \nu - E\delta.$$

The head entrance includes

$$\begin{gathered} 0 < t \le 1, \quad 0 < V < 1, \quad 0 < T, X, Y, Z \le 1, \\ t > V, \quad \tau > 0, \quad \Delta > \tau, \quad \mu > 0, \\ \max\{0, p-1, 1/X - 1, T/Z\} < \beta < \min\{1, V/Y, 5 - f\}, \\ d = tX - Y > 0, \quad tp + d - 1 > 0, \quad 1 + t + d < 33/16. \end{gathered} \tag{B.27}$$

Thus $Q_o = \delta(t - V) + \beta\tau > 0$. Define

$$\begin{aligned} K_{13} &= \frac{p\Gamma + X(V+\beta)}{\Gamma + \beta t X}, \quad K_{14} = \frac{p\mu + X(V+T)}{\mu + X(VE + Tt)}, \\ K_{23} &= \frac{p\Gamma + Y(V+\beta)}{tV}, \quad K_{24} = \frac{p\mu + Y(V+T)}{V(tZ + EY)}. \end{aligned} \tag{B.28}$$

With

$$a_T = 1 - Y + \Gamma(1+X)/\delta, \qquad c_T = 1 + Z + \nu(1+X)/\delta,$$

the endpoint and its complete qualifications are

$$\begin{gathered} E = L_D := \frac{\nu(r-a_T)+Q_o(r-c_T)}{\delta(r-a_T)}, \qquad L_C := \frac{r-1-Z}{1+X}, \\ E > L_C, \quad E < 1, \quad \ell > 0, \quad k = K_{24} > 2, \quad k < K_{13}, K_{14}, K_{23}, \\ b = \frac{\delta(r-a_T)}{Q_o}, \qquad 1+X < b \le k. \end{gathered} \tag{B.29}$$

Here $a_T < 2 < r$, so no denominator in (B.29) vanishes. All four capacity denominators are positive on the declared head. The three strict comparisons in (B.29) are essential.

The five remaining middle-column tests are also joint tests at this same (η, E, k). Set

$$\begin{aligned} A_1 &= 1+Y+t(1-X), & B_1 &= Q_o, \\ C_1 &= 1+Y+t(p-X), & D_1 &= \Gamma + t\beta X, \\ A_2 &= \frac{t(1+Z)+E(1+Y)}{t+E}, & B_2 &= \frac{t\nu - E\Gamma}{t+E}, \\ C_2 &= \frac{tk+E(1+Y)}{t+E}, & D_2 &= \frac{E\Gamma}{t+E}, \qquad H = s/\beta. \end{aligned}$$

Their denominators and B_1, B_2, D_1, D_2 are strictly positive. The tests are

$$\begin{gathered} A_2 + B_2 H \ge r, \qquad A_1 + B_1 s \ge r, \\ \frac{B_2C_1 + D_1A_2}{B_2+D_1}, \quad \frac{B_1C_2 + D_2A_1}{B_1+D_2}, \quad \frac{B_2C_2 + D_2A_2}{B_2+D_2} \ge r. \end{gathered} \tag{B.30}$$

For every head with the declared entrance conditions, the target reduction says that existence of a complete source of height at least f is equivalent to (B.29) and (B.30) at $E = L_D$. A common realization is

$$g = s/V, \quad j = (tk-p)/Y, \quad h = \max\{0, (r-A_1)/B_1, (r-A_2)/B_2\},$$

$$a = (r-1-Y+\Gamma h)/t, \quad v = (b-1-X)/\delta, \quad q = 1-\beta v, \quad w = -r.$$

The proof uses the exact complete-column reconstruction, not separate optimal completions of different columns.

Collecting Lemmas Appendix B.2, Appendix B.4, Appendix B.7, Appendix B.10, and Corollary Appendix B.18, together with the early-pivot bound, gives the necessary inequalities used subsequently:

$$\begin{gathered} 0 < g, j, h \le 1, \quad v, q, \ell > 0, \quad 1 < p < 1+\beta < 2, \quad a < p, \quad k \le 9/4, \\ 0 < t \le 1, \quad t > V, \quad \Delta > \tau, \quad X > 1/(1+\beta), \quad Ek < 1-T, \\ F+\beta < 5, \quad g < 3 - F/2, \quad d > 0, \quad tp + d - 1 > 0, \quad 1+t+d < 33/16. \end{gathered} \tag{B.31}$$

They apply to complete high-value representatives in this positive chamber; they are not imposed as necessary conditions on arbitrary signed X matrices.

Definition Appendix B.22 (Admissible target representative)**.** An admissible target representative is a source in $\mathcal{K}$, at a height $\Lambda \le F < 5$, with all contacts (B.26). In using its high-value necessary inequalities we retain the proved qualifications (B.31). No conditions $t = X = Z = 1$, $b = k$, or $h = s$ are included in this definition.

Appendix B.8. A complete row-scale coordinate system

For an admissible target representative with $\Lambda \le F < 5$, write $F = 2r = 2(1+R)$, $p = 1+s$, and use $(\beta, V, s, Y, Z, v, h, b, R)$ as coordinates. Define

$$\begin{gathered} T = (1+s)Z - 1, \quad q = 1 - \beta v, \\ \Gamma = V - \beta Y, \quad \nu = \beta Z - T, \quad \mu = VZ - TY, \\ A = R + Y + \Gamma v, \qquad C = R - Z + \nu v, \\ W = R(Y+Z) + \mu v, \qquad N = V(Y+Z) + s\mu, \qquad \lambda = \frac{N}{VW}. \end{gathered} \tag{B.32}$$

The complete reconstruction is

$$\begin{gathered} t = A/b, \quad E = C/b, \quad X = (b - 1 - v)/q, \quad k = \lambda b, \\ g = s/V, \quad j = (\lambda A - p)/Y, \quad a = \frac{b(R - Y + \Gamma h)}{A}, \quad w = -r. \end{gathered} \tag{B.33}$$

Inserting these expressions into (B.1) recovers one matrix. In particular the products appearing in different columns retain the same β, p, k, u, x.

To derive the dictionary, the tail contacts give $tb = A$, $Eb = C$, and $b = 1 + v + Xq$. The first-column contacts give $Vg = s$, $tk = p + Yj$, and $Ek + Zj + Tg = 1$. Since $YC + ZA = W$, elimination in these three equations gives $k/b = N/(VW)$. The equation $O_{11} = 1$ then gives the formula for a. These are exact identities with the same variables.

Here is the full domain needed for the converse reconstruction. Define the fixed-coordinate predicate BASE by

$$\begin{gathered} 0 < \beta < 1, \quad 0 < V < 1, \quad 0 < T, Y, Z \le 1, \\ 1 < p < 1 + \beta, \quad F + \beta < 5, \quad 0 < v \le s, \\ \Gamma, \nu, \mu > 0, \quad 0 < g \le 1, \quad \lambda \ge 1, \\ 0 < j < 1 - \beta g, \quad pY - V \le 1, \\ D_h := A\nu - C\Gamma = \mu + R(\nu - \Gamma) > 0, \\ n_h := A(R - Z) + C(R - Y), \qquad 0 < h_0 := n_h/D_h \le H := s/\beta. \end{gathered} \tag{B.34}$$

Throughout this appendix's height range, $R > 1$. Under BASE, $q > 0$, $A > C > 0$, and all denominators in (B.33) are positive once $b \ge A$. Put

$$\begin{gathered} B_0 = 2 + (1 - \beta)v, \qquad B_\tau = 1 + v + qY/V, \\ \sigma_0 = \lambda q - j = \frac{(V + T)(gR - v)}{W} > 0, \qquad B_F = \frac{pq - j(1+v)}{\sigma_0}, \\ I = [A, \infty) \cap (1 + v, \infty) \cap (-\infty, B_0] \cap (-\infty, B_\tau) \cap (-\infty, B_F). \end{gathered} \tag{B.35}$$

The five conditions respectively mean $t \le 1$, $X > 0$, $X \le 1$, $\tau > 0$, and $S_{00} < p$. The interval may be empty or a valid singleton. A singleton at a closed endpoint is retained; a point on either of the last two open endpoints is excluded.

Define

$$\begin{gathered} \mathfrak{a} = R(1+q) + Vv, \qquad \mathfrak{c} = V + \beta R, \\ L(h) = \frac{rA + C(R - Y + \Gamma h)}{A\lambda}, \\ U_O(h) = \frac{A[B_0 + (1+\beta)h]}{\mathfrak{a} + \mathfrak{c}h}, \qquad U_S(h) = \frac{A[pq + (1+v)(1+\beta h)]}{\mathfrak{a} + \mathfrak{c}h}. \end{gathered} \tag{B.36}$$

Proposition Appendix B.23 (Complete scale fibre)**.** *Under* BASE*, a complete representative with the specified fixed coordinates and target height exists if and only if*

$$h_0 \le h \le H, \qquad b \in I, \qquad L(h) \le b \le \min\{U_O(h), U_S(h)\} \tag{B.37}$$

has a solution (b, h)*. Every such solution reconstructs a matrix satisfying all the original physical bands and all six positive cofactors. Its pivots are* $1, p, k, r, -2r$*.*

Proof. The defining identities give

$$A(1 - O_{21}) = D_h h - n_h, \tag{B.38}$$

$$Aq(1 + O_{01}) = A[B_0 + (1 + \beta)h] - b(\mathfrak{a} + \mathfrak{c}h), \tag{B.39}$$

$$Aq(p + S_{01}) = A[pq + (1 + v)(1 + \beta h)] - b(\mathfrak{a} + \mathfrak{c}h). \tag{B.40}$$

The inequality $r + Ea \le k$ is $L(h) \le b$. The receiver inequality $|-1-\beta h| \le p$ is $h \le H$. Together with (B.35), these prove necessity.

For sufficiency, $b \ge A > C > 0$ gives $0 < t \le 1$ and $0 < E < 1$, and $b \in I$ gives $0 < X \le 1$, $\tau > 0$, and $S_{00} < p$. The other cofactors follow from $\Gamma, \nu, \mu > 0$, $\delta = 1 - \beta X > 0$, and $\Delta = Z - TX \ge Z - T > 0$. The signs and the opposite sides of the physical bands are not discarded. They follow as follows:

- The head bounds use $p - \beta \in (0, 1)$, $L_2 = 1$, $L_0 = pX - 1 \in (-1, 1)$, and $L_1 = pY - V \in (-1, 1]$.
- The first receiver column has $0 < g \le 1$, $0 < j < 1 - \beta g < 1 < p$, and $P_0 = -\beta g - j \in (-1, 0)$. Its exact contacts are $S_{10} = p$, $O_{10} = 1$, $O_{20} = -1$. Also $S_{20} = Tg - 1 \in [-1, 0]$. From $k \ge r + Ea > r > 2$ and $Xj < 1$, $S_{00} = k - Xj > 1$. Thus $0 < O_{00} = S_{00} - g < p - g < 1$, using $g = s/V > s$.
- The middle receiver has $0 < h \le H < 1$ and $-p \le \widehat{q}_1 < -1$, with $P_1 = -1$. Equations (B.39)–(B.40) give its two nontrivial lower bands. Their opposite inequalities hold from the reconstructed signs; in particular $S_{01} = -a - X(1 + \beta h) < 0$ and $O_{01} = S_{01} + h < 1$. Furthermore $S_{11} = 1 - Vh \in (0, 1)$, $O_{11} = 1$, and $S_{21} = r + Ea - Z(1 + \beta h) > r - 2 > 0$. Together with $O_{21} = S_{21} + Th \le 1$, this establishes both sides of the remaining middle-column bands.
- The tail has $P_2 = 1$, $(O_{02}, O_{12}, O_{22}) = (-1, 1, -1)$, and

  $$(S_{02}, S_{12}, S_{22}) = (-1 - v, 1 - Vv, -1 - Tv).$$

  The condition $v \le s = p - 1$ establishes all three p-bands.
- The core bounds use $a < p < 2 < k$, $b \le k$, $tk \le k$, $Ek < k$, and $r + Ea \le k$. Here $a > 0$ follows from $R > 1 \ge Y$, and $a < p$ from $S_{01} \ge -p$. The remaining entries satisfy $D_{11} = 1 + Y - \Gamma h \in (0, 2)$, $D_{12} = 1 - Yq - Vv \in (-1, 1)$, and $D_{22} = -r + C \in (-r, 0)$.

All entries are those of the same reconstructed matrix. These identities establish the complete bands, not only a subset of necessary inequalities. They also recover the strict first-column capacity comparisons in (B.29) from the first-column residual identities and $S_{00} < p$, $P_0 > -1$. □

Why no division or endpoint is hidden.. The interval I in (B.35), rather than its closure, is used in the equivalence. The quantities $V, Y, q, A, \lambda, D_h, \sigma_0, \mathfrak{a} + \mathfrak{c}h$ are positive in their stated domains. A zero denominator outside this domain is not an infeasibility proof: such a point has not met the declared qualification. Later, on the certified near-peak box, these qualifications have uniform margins, so its closure does not introduce zero-cofactor or excluded-endpoint cases.

Appendix B.9. Common necessary inequalities and the finite localization

Put $\epsilon = R - 1 > 0$. Two slacks of the same physical completion have numerators

$$\mathcal{T} = (s - Vv)\mu - \epsilon V(Y + Z) = VW(\lambda - 1) \geq 0,$$

$$\mathcal{P} = \epsilon V[(Y + Z) - v(\nu - \Gamma)] - (s - Vv)D_h = VW(1 - j - \beta g) > 0.$$

Their sum is

$$\mathcal{T} + \mathcal{P} = (sR - Vv)(\Gamma - \nu). \tag{B.41}$$

Because $v \leq s$ and $V < 1 < R$, the first factor on the right is positive. Therefore $\Gamma > \nu$ throughout the qualified high-value completion domain. This is a common-slack identity, not a sign inferred from the candidate matrix.

The two middle capacities satisfy

$$U_O - U_S = \frac{Aq(h - s)}{\mathfrak{a} + \mathfrak{c}h}, \qquad U_O' > 0, \quad U_S' < 0, \quad L' > 0.$$

For example

$$U_O' = \frac{2Aq(R - V)}{(\mathfrak{a} + \mathfrak{c}h)^2}, \qquad U_S' = -\frac{Aq[\beta R(p - 1 - v) + V(p + 1 + v)]}{(\mathfrak{a} + \mathfrak{c}h)^2}.$$

It follows that every feasible (b, h) satisfies $A \leq b \leq U_O(s) = U_S(s)$, without setting $h = s$. Define

$$\begin{gathered} d_1 = 1 - V - \beta\epsilon, \quad a_h = \epsilon(1 + q) - (1 - V)v, \\ d_S = V + \beta(\epsilon - v), \quad s_h = pq + 1 + v - \mathfrak{a}, \\ h_A = a_h/d_1, \qquad h_S = s_h/d_S. \end{gathered} \tag{B.42}$$

Feasibility forces $d_1 > 0$: indeed

$$B_0 + (1 + \beta)h - \mathfrak{a} - \mathfrak{c}h = d_1(h + v) - 2\epsilon q$$

and $b \geq A$, $b \leq U_O(h)$ make the left side nonnegative. Also $d_S > 0$, since $v \leq V$, $\beta < 1$, and $\epsilon > 0$. The two middle inequalities then imply $h_A \leq h \leq h_S$. Equivalently, with

$$\mathcal{M} = (1 - V)(s + v) - \epsilon[2 + \beta(s - v)],$$

one has $\mathcal{M} \geq 0$, $h_A = s - \mathcal{M}/d_1$, and $h_S = s + \mathcal{M}/d_S$.

For reference we specify the necessary system used in the finite localization. This makes the connection from the physical domain to its rational certificate explicit. In the following table all expressions are evaluated at the same six coordinates (β, V, s, Y, Z, v) and the same fixed R; they are not independently selected bounds. In addition to the definitions above, put

$$u_0 = rA + C(R - Y), \quad u_1 = C\Gamma, \quad B_s = B_0 + (1 + \beta)s, \quad D_s = \mathfrak{a} + \mathfrak{c}s,$$

$$\begin{aligned} c_0 &= NA^2B_0 - VWu_0\mathfrak{a}, \\ c_1 &= NA^2(1 + \beta) - VW(u_0\mathfrak{c} + u_1\mathfrak{a}), \\ c_2 &= -VWu_1\mathfrak{c} < 0. \end{aligned}$$

Thus $c_0 + c_1h + c_2h^2 = VW\Pi(h)$, where

$$\Pi(h) = A^2\lambda[B_0 + (1 + \beta)h] - (u_0 + u_1h)(\mathfrak{a} + \mathfrak{c}h).$$

The unconditional necessary inequalities are $g_i \geq 0$, $0 \leq i \leq 34$, with the following exact polynomial dictionary.

i	g_i
0–6	$\beta - s,\ T,\ \nu,\ \Gamma,\ \mu,\ s - v,\ V - s$, in this order.
7	$B_0 - A$.
8	$(s - Vv)\mu - \epsilon V(Y + Z)$.
9	d_1.
10	$(1 - V)(s + v) - \epsilon[2 + \beta(s - v)]$.
11	D_h.
12	$\beta s - s^2 - \epsilon\beta$.
13	$3 - 2R - \beta$.
14	$(1 + \beta)Y - V$.
15	$1 - pY + V$.
16	$sD_h - \beta n_h$.
17	$s_h D_h - n_h d_S$.
18	$NAB_0 D_h - VW(u_0 D_h + u_1 n_h)$.
19	$NAB_0 d_1 - VW(u_0 d_1 + u_1 a_h)$.
20	$\mathcal{P}$.
21	$Nq - (V + T)(sR - Vv)$.
22	$pqVW - NqA + [Nq - (V + T)(sR - Vv)](A - 1 - v)$.
23	$qY + V(1 + v) - VA$.
24	$NA^2 B_s D_h - VWD_s(u_0 D_h + u_1 n_h)$.
25	$AB_s - (1 + v)D_s$.
26	$c_1^2 - 4c_0 c_2$.
27	$9VW - 4NA$.
28	$(9A - 4u_0)D_h - 4u_1 n_h$.
29	$(1 - T)VW - CN$.
30	$(3 - r)V - s$.
31	$17q - 16R - 16Vv + 16v(1 + \beta)$.
32	$B_0(R + Vv) - A(1 + v)$.
33	$(R + Vv - q)B_0 + A(pq - 1 - v)$.
34	$\Gamma - \nu$.

The table is a necessary system, not the definition of a complete physical source. Its rows follow by clearing positive denominators in the scale conditions, the common bounds on h, and (B.31). To indicate all distinct sources: rows 0–17 use BASE, $A \le b \le B_0$, the tail capacity, and $h_0 \le H, h_S$; rows 18–19 use $L(h_0), L(h_A) \le B_0$; rows 20–23 use the strict first-column bands and $\tau > 0$. Rows 24–25 use the shared middle bound at s, row 26 the discriminant of the same concave quadratic, rows 27–30 the original pivot and head budgets, and rows 31–33 the three original geometric entrance inequalities. For the latter, the identities are

$$d = \frac{R + Vv - A(1 + v)/b}{q}, \quad 1 + t + d = 1 + \frac{R + Vv}{q} - \frac{Av(1 + \beta)}{qb},$$

$$tp + d - 1 = \frac{(R + Vv - q)b + A(pq - 1 - v)}{bq}.$$

The bounds $A \le b \le B_0$ give the displayed necessary rows with their stated directions. Row 34 is (B.41).

There is one conditional row, not an additional unconditional assumption:

$$g_{36} := c_1 + 2c_2 s \ge 0 \quad \Longrightarrow \quad g_{35} := c_0 + c_1 s + c_2 s^2 \ge 0. \tag{B.43}$$

If a feasible $h \le s$ exists, the nonnegative derivative at s makes the concave quadratic increasing up to s, so $\Pi(s) \ge \Pi(h) \ge 0$. If $h \ge s$, then $L(s) \le L(h) \le U_S(h) \le U_S(s) = U_O(s)$, with

the same conclusion. Any certificate row using g_{35} must first establish $g_{36} \geq 0$ on its entire current box.

Set

$$f_0 = \frac{4132517}{10^6}, \quad f_1 = \frac{103313}{25000}, \quad R_0 = \frac{2132517}{2000000}, \quad R_1 = \frac{53313}{50000}. \tag{B.44}$$

The root enclosure for height f_0 is

$$\mathcal{B}_0 = [1/4, 7/8] \times [1/16, 1] \times [1/16, 7/8] \times [0, 1] \times [1/2, 1] \times [0, 7/8]. \tag{B.45}$$

Indeed the common middle and core bounds give

$$\epsilon + Ea \leq \frac{T(\beta - s)}{\beta} \leq \frac{s(\beta - s)}{\beta} \leq \frac{\beta}{4}.$$

Thus $s > \epsilon > 1/16$ and $\beta > 4\epsilon > 1/4$. Also $s \leq V < 1$, $v \leq s < \beta < 5 - f_0 < 7/8$, and $Z > 1/(1 + \beta) > 1/2$. Hence every relevant six-tuple lies in $\mathcal{B}_0$. Its zero endpoints are only outer relaxations.

Proposition Appendix B.24 (Finite localization and upper-target exclusion)**.** *At $R = R_0$, every point of $\mathcal{B}_0$ satisfying the 35 unconditional inequalities and the conditional implication* (B.43) *lies in*

$$\begin{aligned} Q = {} & [0.617, 0.6181] \times [0.7788, 0.7795] \times [0.4525, 0.4540] \\ & \times [0.97352, 0.97361] \times [0.99998, 1] \times [0.1944, 0.1950]. \end{aligned} \tag{B.46}$$

All finite decimals here and below denote exact rational numbers. At $R = R_1$, the same necessary system is infeasible in the box obtained from Q by replacing its last interval by $[0.1944, 0.19501]$.

Finite proof objects and acceptance rule.. The first statement is the repaired complete localization certificate: 152,467 records, including 35,069 binary splits, 82,328 contraction nodes, 3,338 direct contradiction terminals, 31,729 nonnegative combination terminals, and three terminals wholly contained in Q. The contraction nodes contain 208,274 individually checked thin-slice exclusions. The second statement is the repaired upper-target tree, with 169 nodes and 85 contradiction terminals. Neither statement uses the earlier, superseded localization or upper-target tree counts.

Their small mathematical acceptance rule is as follows. On a current box use exact outward rational derivative and Hessian enclosures to obtain affine upper bounds $g_j(x) \leq b_j + \sum_i a_{ji} z_i$, $|z_i| \leq 1$. If $g_j \geq 0$, the resulting affine row is necessary. Nonnegative rational weights w_j produce a contradiction when

$$\sum_j w_j b_j + \sum_i \left| \sum_j w_j a_{ji} \right| < 0.$$

The same inequality applied to an edge slice justifies a contraction. Both children of each binary split cover their parent; the contracted child contains every previously possible point. At a localization terminal, containment of the entire current box in Q is checked coordinate by coordinate. This proves the statement for continuous real boxes; the rational mesh records enclosure endpoints, not a discrete sample of possible source points. The complete finite payloads and entry points are identified in §Appendix B.13.

Appendix B.10. Transport of the target and a compact nine-variable model

The fixed-target statement alone would not suffice. Fix the old head η of a high representative. Its tail identities give

$$tb = R + Y + \Gamma v, \qquad b = 1 + X + (1 - \beta X)v,$$

so

$$v(R) = \frac{R + Y - t(1 + X)}{Q_o}, \qquad Q_o = t(1 - \beta X) - \Gamma > 0. \tag{B.47}$$

Choose its height-f_0 representative using Proposition Appendix B.21. Its six-tuple lies in Q. Since $b \geq A$, $b \leq B_0$, and $X \leq 1$,

$$Q_o \geq (A/B_0)(1 - \beta) - \Gamma > 1/5.$$

The last strict inequality is a rational enclosure on the entire height-f_0 box, not just its center. It follows that

$$0 \leq v(R) - v(R_0) \leq 5(R_1 - R_0) = \frac{3}{400000} < 10^{-5} \qquad (R_0 \leq R \leq R_1). \tag{B.48}$$

The coordinates β, V, s, Y, Z do not change with the target, since $s = (1+T)/Z - 1$ is fixed by the old head. If a source had height at least f_1, it would have a target representative at f_1 with this same head. Equations (B.47)–(B.48) would place it in the upper-target exclusion box, a contradiction. This proves $F < f_1$ for every high representative, including those initially specified above f_1. The comparison is between representatives at two targets; it does not assert existence of a continuous physical flow in E.

The bounds $A \leq b \leq B_0$ and $h_A \leq h \leq h_S$ now put every height-$F \geq f_0$ completion in

$$\begin{aligned} \mathcal{B} = Q_{\beta,V,s,Y,Z} &\times [0.1944, 0.19501]_v \times [0.4513, 0.4566]_h \\ &\times [2.0741, 2.0748]_b \times [R_0, R_1]_R. \end{aligned} \tag{B.49}$$

The relevant verified margins and enclosure statements are gathered in the next proposition so that the inverse-function argument has an explicit physical domain.

Proposition Appendix B.25 (Uniform qualifications)**.** *Throughout $\mathcal{B}$, the rational expressions in* (B.32)–(B.42) *have positive denominators, and the following strict inequalities hold:*

$$\begin{gathered} q, \Gamma, \nu, \mu, A, C, W, N, D_h, d_1, d_S, j, g, T > 0, \\ T < 1, \quad g < 1, \quad E < 1, \quad 0 < v < s < \beta < 1, \quad h < H, \\ 1 - j - \beta g > 1/1000, \qquad p - S_{00} > 1/200, \\ pY - V < 1, \quad F + \beta < 5, \quad X > 0, \quad b > 1 + v, \\ 0 < \beta, V, Y < 1. \end{gathered}$$

For any point in this box with $X \leq 1$, also

$$\delta \geq 1 - \beta > 0, \quad \tau \geq Y - V > 0, \quad \Delta \geq Z - T > 0.$$

In the enlarged six-coordinate box and height interval, the same rational enclosures give

$$A > 2.0741, \quad B_0 < 2.0748, \quad h_A > 0.4513, \quad h_S < 0.4566.$$

At the low target the transport denominator has the lower bound $(A/B_0)(1 - \beta) - \Gamma > 1/5$.

These are 42 finite rational interval checks, including the matching of the nine-dimensional Jacobian box with (B.49). The expression for $1 - j - \beta g$ is enclosed using $\mathcal{P}/(VW)$; this preserves the common-slack cancellation in (B.41). The three last cofactor lower bounds are conditional on $X \leq 1$, which will be supplied by an explicit nonnegative slack, rather than being claimed on the whole outer box.

Appendix B.11. Eight physical slacks and a uniform Jacobian

On the domain of the reconstruction define the following eight slacks, all with the feasible direction $G_i \geq 0$:

$$\begin{aligned} G_1 &= b - A, & G_2 &= B_0 - b, \quad G_3 = \lambda - 1, \\ G_4 &= \lambda b - r - \frac{C(R - Y + \Gamma h)}{A}, \\ G_5 &= A[B_0 + (1+\beta)h] - b(\mathfrak{a} + \mathfrak{c}h), \\ G_6 &= A[pq + (1+v)(1+\beta h)] - b(\mathfrak{a} + \mathfrak{c}h), \\ G_7 &= D_h h - n_h, & G_8 &= 1 - Z. \end{aligned} \tag{B.50}$$

Their physical meanings, in the same order, are

$$\begin{gathered} t \leq 1, \quad X \leq 1, \quad b \leq k, \quad r + Ea \leq k, \\ O_{01} \geq -1, \quad S_{01} \geq -p, \quad O_{21} \leq 1, \quad Z \leq 1. \end{gathered}$$

For example $G_1 = b(1-t)$, $G_2 = q(1-X)$, and $G_3 = (k-b)/b$; the identities (B.38)–(B.40) give $G_5 = Aq(1+O_{01})$, $G_6 = Aq(p+S_{01})$, and $G_7 = A(1-O_{21})$. All these factors are positive.

Lemma Appendix B.26 (Exact feasible set in the localized box)**.** *Let $\mathcal{C}$ be the set of all complete scale representatives at heights $F \geq f_0$, with the source and qualifications of Propositions Appendix B.21 and Appendix B.23. Then*

$$\mathcal{C} = \{z \in \mathcal{B} : G_i(z) \geq 0 \ (1 \leq i \leq 8)\}. \tag{B.51}$$

In particular, $\mathcal{C}$ is compact and every member has $F < f_1$.

Proof. The forward inclusion is the localization and target-transport argument above, followed by the physical interpretation of the slacks. For the reverse inclusion, all the strict constant qualifications in BASE follow from Proposition Appendix B.25; its non-strict tail capacity $\lambda \geq 1$ is $G_3 \geq 0$. To check explicitly the only remaining condition involving h_0, observe that $R > 1$, $Y < 1$, and $Z \leq 1$ imply $n_h > 0$. Thus $G_7 \geq 0$ gives $0 < h_0 \leq h < H$, using $D_h > 0$. The constraints $G_1, G_2 \geq 0$ give $A \leq b \leq B_0$. The other endpoints of I follow from $X > 0$, $\tau > 0$, and $p - S_{00} > 1/200$; the cofactor bound for τ here uses precisely $G_2 \geq 0$. Hence $b \in I$, with both strict endpoints respected. Finally $G_4, G_5, G_6 \geq 0$ are exactly $L(h) \leq b \leq \min\{U_O(h), U_S(h)\}$. Proposition Appendix B.23 now supplies every original physical band of one complete matrix. The inherited high-value necessary bounds apply to this complete representative as specified in §Appendix B.13.

All eight functions are continuous on a neighborhood of the closed box because their denominators are bounded away from zero there. The right side of (B.51) is therefore a closed subset of a compact box. No previously strict cofactor or first-column condition has been replaced by a boundary case. The independent upper-target exclusion already proves $F < f_1$ on this set. □

Write $\zeta = (\beta, V, Y, Z, v, h, b, R)$, so that $z = (\beta, V, s, Y, Z, v, h, b, R)$ and $J = \partial(G_1, \dots, G_8)/\partial\zeta$ at fixed s.

Proposition Appendix B.27 (Uniform slack coordinates)**.** *For every $z \in \mathcal{B}$, $J(z)$ is invertible. In the local coordinates $(s, G_1, \dots, G_8)$,*

$$\frac{\partial R}{\partial G_i} < -\frac{1}{250} \qquad (1 \leq i \leq 8). \tag{B.52}$$

Finite rational verification of the proposition.. The repaired certificate specifies a fixed rational matrix C_0, a rational vector y_0, and positive rational radii ρ_i. With the rational midpoint of $\mathcal{B}$ as center, exact interval Hessians enclose every entry of J by its central value plus its variation over all nine coordinates. The verifier checks, for the entire box,

$$\mathcal{E} = I - C_0^T J^T, \qquad \|\mathcal{E}\|_\infty < \frac{7}{50} < 1. \tag{B.53}$$

Thus $C_0^T J^T$, and hence J, are invertible. Let $e_R = (0,0,0,0,0,0,0,1)^T$. The same verifier bounds the centered residual $C_0^T(e_R - J^T y_0)$ and proves componentwise that

$$\left|C_0^T(e_R - J^T y_0)\right|_i + \sum_j |\mathcal{E}_{ij}|\rho_j < \rho_i.$$

Consequently the affine map $y \mapsto y + C_0^T(e_R - J^T y)$ maps $y_0 + [-\rho, \rho]$ strictly into itself, uniformly in z. Its unique fixed point solves $J^T y = e_R$, so $y_i = \partial R/\partial G_i$. In the stated order the exact upper bounds $y_{0,i} + \rho_i$ are

$$-\frac{31023639}{10^9}, \quad -\frac{184717889}{10^9}, \quad -\frac{47423419}{500000000}, \quad -\frac{5888171}{25000000},$$
$$-\frac{380657}{25000000}, \quad -\frac{834537}{62500000}, \quad -\frac{9580477}{200000000}, \quad -\frac{4195499}{500000000}.$$

Each is strictly smaller than $-1/250$. These are interval conclusions on the full nine-variable box, not gradients at its center or a finite set of sampled points. The precise certificate data and verifier are identified in §Appendix B.13.

Lemma Appendix B.28 (A supercritical maximum has all eight contacts)**.** *If $\mathcal{C}$ contains a point of height greater than α, it has a maximizing point z_* with $f_0 < F(z_*) < f_1$, and every such maximizing point satisfies $G_1(z_*) = \cdots = G_8(z_*) = 0$.*

Proof. Compactness gives a maximum. Under the stated hypothesis its height is greater than $\alpha > f_0$; the upper-target exclusion makes it less than f_1. This existence argument does not require an additional numerical or rational witness.

Suppose $G_i(z_*) > 0$. The inverse-function theorem, using Proposition Appendix B.27, supplies a local smooth inverse in the eight slack coordinates at fixed s. Decrease this one slack by a positive amount u, leaving the other seven fixed. For sufficiently small u, all eight slacks remain nonnegative. At $u = 0$, (B.52) gives $dR/du > 1/250$, so a short arc strictly increases height. All the other strict qualifications persist by continuity. The reconstruction theorem therefore makes this a feasible arc in the original complete scale model, with height still above f_0.

The artificial faces of $\mathcal{B}$ are not additional physical constraints. In constructing this short arc one works in an open neighborhood on which the reconstruction denominators and strict qualifications stay positive, and imposes only the eight physical slacks. If a point of the arc appeared to leave $\mathcal{B}$, the already proved global localization, applied to that original feasible point, would force it back into $\mathcal{B}$. Thus the arc belongs to $\mathcal{C}$, contradicting maximality. This proves all eight equalities. □

Appendix B.12. The contact carrier and its exact elimination

The contact family is a consequence of the preceding maximum argument. Indeed $G_1 = G_2 = G_8 = 0$ gives $t = X = Z = 1$, whence $T = s$. The equations $G_3 = G_4 = 0$ give $b = k = r + Ea$. Also

$$G_5 - G_6 = Aq(h - s),$$

so $h = s$, and the last middle contacts are $O_{01} = -1$, $S_{01} = -p$, and $O_{21} = 1$. These contacts use the original shared matrix variables.

Put $\epsilon = r - 2 = R - 1$, and let $\mathcal{L} = (2 - s)(1 + s)$. We give the elimination from these contacts to the carrier explicitly. The top middle contact gives $a = (1 - \beta)s$; the bottom middle contact gives $\epsilon + Ea = s(\beta - s)$. Combining the latter with $b = r + Ea$ and the top tail contact gives

$$b = 2 + s(\beta - s), \qquad (1 - \beta)v = s(\beta - s).$$

The bottom tail contact is $Eb = \epsilon + (\beta - s)v$. Substitute these expressions into $(b - r)b = a(Eb)$. After clearing only $1 - \beta > 0$, the equation reduces to $2s(\beta - s) = \epsilon\mathcal{L}$. Thus

$$\begin{gathered} \beta = s + \frac{\epsilon\mathcal{L}}{2s}, \qquad v = \frac{s(\beta - s)}{1 - \beta}, \qquad b = 2 + s(\beta - s), \qquad a = (1 - \beta)s, \\ Y = 1 - \frac{\epsilon(s - v)}{s + v}, \qquad V = 1 - \frac{\epsilon[2 + \beta(s - v)]}{s + v}. \end{gathered} \tag{B.54}$$

For completeness, the last two formulas are the solution of the middle and tail receiver equations

$$R - a = Y(1 + \beta s) - Vs, \qquad b - R = Yq + Vv.$$

Their determinant is $v(1 + \beta s) + sq = s + v > 0$. The remaining first-column relation, from $Eb + Zj + Tg = 1$, $j = (b - p)/Y$, and $g = s/V$, is

$$\epsilon + (\beta - s)v + \frac{b - 1 - s}{Y} + \frac{s^2}{V} = 1. \tag{B.55}$$

To specify a compact integer polynomial, define

$$\begin{gathered} d_* = 2s(1 - s) - \epsilon\mathcal{L}, \qquad B_* = 2s^2 + \epsilon\mathcal{L}, \\ Y_n = s(1 - s)(1 - \epsilon) + \epsilon^2\mathcal{L}, \qquad Y_d = s(1 - s), \\ V_n = 2s^2(1 - s) - 2\epsilon d_* - \epsilon B_* s(1 - s) + \epsilon^2 B_*\mathcal{L}, \qquad V_d = 2s^2(1 - s). \end{gathered}$$

The subscript on d_* distinguishes it from the geometric entrance quantity $d = tX - Y$. Direct substitution in (B.54) gives $Y = Y_n/Y_d$ and $V = V_n/V_d$. Define

$$\begin{aligned} G(s, \epsilon) = {} & [2d_*(1 - \epsilon) - \epsilon^2\mathcal{L}^2]Y_n V_n \\ & - d_*(2 - 2s + \epsilon\mathcal{L})Y_d V_n - 4d_* s^4(1 - s)Y_n. \end{aligned} \tag{B.56}$$

This polynomial has degree 10 in s and degree 7 in ϵ. With β, v, b, Y, V given by the same carrier dictionary, its exact relation to the residual is

$$\epsilon + (\beta - s)v + \frac{b - 1 - s}{Y} + \frac{s^2}{V} - 1 = -\frac{G(s, \epsilon)}{2d_* Y_n V_n}. \tag{B.57}$$

Here $0 < s < \beta < 1$, $d_* = 2s(1 - \beta) > 0$, $Y_n = YY_d > 0$, and $V_n = VV_d > 0$. Thus no zero-denominator component has entered the carrier equation: (B.55) is equivalent to $G(s, \epsilon) = 0$ in the proved physical domain.

Lemma Appendix B.29 (Stationarity on the attained contact curve)**.** *At the supercritical maximizing point of Lemma Appendix B.28,*

$$G(s, \epsilon) = 0, \qquad G_s(s, \epsilon) = 0.$$

Proof. The same invertible Jacobian makes all eight coordinates ζ smooth functions of s when the eight slacks are held at zero. For s in a two-sided open interval about its maximizing value, all strict physical qualifications persist. The contact equalities preserve the eight physical constraints, and the reconstruction therefore gives a curve of complete physical representatives. As in Lemma Appendix B.28, global localization handles any artificial face of the recorded box. Since the maximum is strictly above f_0, neither an auxiliary height endpoint nor an original physical endpoint prevents this two-sided curve. Maximality gives $\epsilon'(s) = R'(s) = 0$. Differentiating the identity $G(s, \epsilon(s)) = 0$ now gives $G_s = 0$. This argument does not assume $G_\epsilon \neq 0$; the full eight-slack Jacobian has already supplied the local parameter. □

Proposition Appendix B.30 (Exact elimination identity)**.** *For the polynomial* (B.56) *and the normalization of* P_{61} *in (M2),*

$$\mathrm{Res}_s(G, G_s) = \frac{\epsilon^{34}(\epsilon-2)(\epsilon-1)^2(\epsilon+1)^7(\epsilon^2-2\epsilon+2)^2}{131072} \cdot P_{61}(4+2\epsilon). \tag{B.58}$$

This is a polynomial identity over $\mathbb{Q}$, checked by computing the resultant and verifying that the cleared polynomial difference is identically zero. Its degree in ϵ is 109; the factor outside P_{61} has degree 48. The carrier dictionary and (B.57) are checked separately, so the identity is connected to the complete physical maximum before root identification is used.

At the maximum,

$$\frac{132517}{2000000} < \epsilon < \frac{3313}{50000}.$$

Every factor outside P_{61} in (B.58) is nonzero there. The two equations of Lemma Appendix B.29 imply that the resultant vanishes, and therefore $P_{61}(F) = 0$. Section M2 has already proved that α is the unique root of this polynomial in $(4, 5)$, which contains $[f_0, f_1]$. Consequently the maximizing height must be α, contradicting the assumed supercritical point. The representative reduction then proves Theorem M6.9 for every source in $\mathcal{K}$. Neither a stationary-point calculation nor the root-isolation certificate alone supplies this global bound.

Appendix B.13. Source dependencies and the finite proof payloads

The analytic representative and reconstruction arguments are proved in §Appendix B.2–§Appendix B.7 above. The historical labels in the following crosswalk identify their sources and the frozen finite certificates still used by those proofs. The source-to-model arguments, the exact certificate acceptance, and the final composition have separate responsibilities; a localization tree does not by itself establish the preceding representative theorems.

Responsibility	Proof in this article and archived origin
High-value signs and quantitative budgets	Lemma Appendix B.2 and Lemma Appendix B.4; V11/V13 safe-tail argument and V14 budgets.
Common first-column reconstruction	Lemmas Appendix B.6–Appendix B.8; V15 and V19.
Simultaneous middle column	Lemma Appendix B.9; V15/V18, with one shared head, row scale and pivot.
Tail polygon and wall reconstruction	Lemmas Appendix B.10–Appendix B.14; V16–V19, including zero-scale and boundary qualifications.
Same-head wall maximum	Theorem Appendix B.16; V20, with the separate finite C19 exclusion of Proposition Appendix B.15.
Head entrance	Proposition Appendix B.17 and Corollary Appendix B.18; V22's finite difference exclusion and analytic reconstruction.
Exceptional contacts and prescribed target	Proposition Appendix B.19, Lemma Appendix B.20, and Proposition Appendix B.21; V23/V24.
Row-scale equivalence	Proposition Appendix B.23; V25.
Localization and exact canonical height	The later sections of this appendix; the repaired V28 localization, transport, peak-neighborhood and algebraic modules.

The finite representative exclusions have three distinct scopes: eight C19 systems in Proposition Appendix B.15, eight difference systems in Proposition Appendix B.17, and six exceptional-contact systems in Proposition Appendix B.19. Their continuous source coverage is derived above before their accepted trees are used. The safe-tail polynomial positivity checks are a separate fixed finite certificate. The Supplementary Index, S3, identifies the nested V24 replay with its full-chain option and the recorded acceptance of all 22 exclusion trees. Neither the V28 localization entry nor the final global composer re-executes these upstream trees automatically.

The reproducibility unit for the localization and peak-height statements in the later part of this appendix is `rho5_v28_exact.zip`, with internal root `rho5_v28_exact/`. Its top-level report, Sections 1–5, specifies the repaired localization and its interfaces. The `RHO5_V26_REBUILT/REPORT.md` retained within it supplies the mathematical argument in its Theorem 6.1 and Sections 7–10. The latter report's superseded tree counts and old Jacobian contraction bound are not the data used in this appendix.

The order of the mathematical dependencies is essential:

1. The same-head wall maximum and the simultaneous reconstruction first give a complete representative at the prescribed attainable height. Compactness and minimization in E are applied to the complete fixed-head fiber. Only then do the exceptional-contact exclusions and the strict capacity comparisons force $E = L_D$, $k = K_{24}$, and the five simultaneous middle-column tests. These steps are proved in Proposition Appendix B.21; the V25 handoff is a historical restatement, not an additional premise.
2. The high-value bounds (B.31) are used only in their declared positive canonical domain. The common source budgets and Lemma Appendix B.7 supply $Ek < 1 - T$, $F + \beta < 5$, and $g < 3 - F/2$ before wall selection; the latter becomes $g < 3 - r$ on the wall. The earlier-pivot bound supplies $k \leq 9/4$. Proposition Appendix B.17 and

Corollary Appendix B.18 supply the three entrance inequalities used in localization rows 31–33. These conclusions are not inferred from an arbitrary signed matrix or from an observed extremal candidate.

3. The scale dictionary, BASE, strict scale endpoints, and both directions of the complete fiber equivalence are proved in §Appendix B.8. The archived V25 report gives their original derivation and exact symbolic checks. This appendix does not use its later finite elimination of β.
4. The repaired V28 certificate supplies continuous root-box coverage, fixed-target transport margins, the upper-target exclusion, the nine-dimensional slack theorem, and the exact carrier/resultant connection. None uses the complete X-domain or negative-diagonal bound as a premise. Root uniqueness and the definition of the exact constant are the separate result of Section M2.

The Supplementary Index identifies the complete replay entry point and the separate localization, symbolic-interface, target-transport, Jacobian, carrier, and resultant checks. Acceptance requires every localization graft and all required modules. Discovery output or summary counts do not replace that finite replay. The representative reductions and their explicit finite inputs have distinct domains: the canonical-height certificate alone does not prove the other sign chambers, degenerate entrances, or the full matrix theorem.

Appendix C. Data and coordinates for the two-gap transport

Appendix C.1. Complete coordinates and source inequalities

The coordinates of ξ have the order in (M63). All entries of the vectors u, x, v, q have indices $0, 1, 2$. Matrices D, S, O below use these same indices. The scalar coordinates A, B are entries of D, not the original input matrix. Define

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & r - \sigma + cB \\ dk & r - \tau + dA & w + dB \end{pmatrix}, \qquad S = D + xq^T, \qquad O = S + uv^T. \tag{C.1}$$

The associated real 5×5 matrix is

$$M(\xi) = \begin{pmatrix} 1 & -e & v^T \\ \beta & p - e\beta & (q + \beta v)^T \\ u & px - eu & O \end{pmatrix}. \tag{C.2}$$

Its successive leading Schur matrices, before the last scalar, are

$$\begin{pmatrix} p & q^T \\ px & S \end{pmatrix}, \qquad D, \qquad \begin{pmatrix} r & r - \sigma \\ r - \tau & w \end{pmatrix}. \tag{C.3}$$

These are algebraic identities for $p, k \neq 0$; elimination at the fourth pivot additionally uses $r \neq 0$. In particular, changing the source coordinates in the transport changes one complete matrix through (C.2), with all its stage bounds retained.

For a scalar expression f with band $|f| \leq b$, write $f^+ = b - f$ and $f^- = b + f$. The superscripts in the tables below always denote these slacks, not the positive and negative parts of a number. The full polynomial list consists of the following bands, with the identically zero slack $D_{00}^+ = k - k$ omitted:

$$\begin{gathered} e^\pm = 1 \mp e, \quad \beta^\pm = 1 \mp \beta, \quad \mathrm{head}^\pm = 1 \mp (p - e\beta), \\ u_i^\pm = 1 \mp u_i, \quad x_i^\pm = 1 \mp x_i, \quad v_i^\pm = 1 \mp v_i, \quad q_i^\pm = p \mp q_i, \\ L_i^\pm = 1 \mp (px_i - eu_i), \qquad P_i^\pm = 1 \mp (q_i + \beta v_i), \\ D_{ij}^\pm = k \mp D_{ij}, \qquad S_{ij}^\pm = p \mp S_{ij}, \qquad O_{ij}^\pm = 1 \mp O_{ij} \qquad (0 \leq i, j \leq 2). \end{gathered} \tag{C.4}$$

There are 95 nonzero polynomials in (C.4). The remaining nine are

$$r + w, \quad r - w, \quad p, \quad k, \quad r, \quad \sigma, \quad \tau, \quad r - \sigma, \quad r - \tau. \tag{C.5}$$

All 104 are required to be nonnegative for a physical source, with $p, k, r > 0$. On the boxes used here the strict positivity of these three pivots follows from the stronger bounds below. These conditions give $\|M\|_{\max} = 1$, successive stage maxima p, k, r, and legal leading complete pivots $1, p, k, r$, by (C.3). Since $w < 0$ on these boxes, the last pivot is $-F$ with $F > 0$. No strict inequality between competing maximal entries is required, so tied pivots are included.

For each j, the components of $\Phi_{j,p}$ are the twenty slacks in Table C.3, in its displayed order, followed by $r + w, \sigma, \tau$. The row index in that table is the index within this selected list; it is not a numbering of all 104 source polynomials. The Jacobian columns have the order

$$\begin{aligned} y = (&k, r, w, A, B, c, d, e, \beta, u_0, u_1, u_2, x_0, x_1, x_2, \\ &v_0, v_1, v_2, q_0, q_1, q_2, \sigma, \tau). \end{aligned} \tag{C.6}$$

Thus J_j is a 23×23 matrix. Its entries remain polynomials in all 24 coordinates, including the parameter p.

Table C.3: The twenty selected source slacks in each local chart.

Index	$j = 0$	$j = 1$	$j = 2$	$j = 3$
0	e^+	e^+	e^+	e^+
1	u_0^+	u_0^+	u_0^+	u_0^+
2	x_0^+	x_0^+	x_0^+	x_0^+
3	S_{01}^-	D_{01}^-	S_{01}^-	D_{01}^-
4	O_{01}^-	O_{01}^-	O_{01}^-	O_{01}^-
5	D_{02}^-	S_{02}^-	D_{02}^+	S_{02}^+
6	O_{02}^-	O_{02}^-	O_{02}^+	O_{02}^+
7	P_1^-	P_1^+	x_1^+	x_1^-
8	D_{10}^+	D_{10}^+	L_1^+	L_1^-
9	S_{10}^+	S_{10}^+	P_1^-	P_1^+
10	O_{10}^+	O_{10}^+	O_{10}^-	O_{10}^+
11	O_{11}^+	O_{11}^+	D_{11}^+	O_{11}^+
12	O_{12}^+	O_{12}^+	O_{11}^+	D_{12}^+
13	x_2^+	x_2^-	O_{12}^+	O_{12}^+
14	L_2^+	L_2^-	P_2^-	P_2^+
15	P_2^+	P_2^-	D_{20}^+	D_{20}^+
16	O_{20}^-	O_{20}^+	S_{20}^+	S_{20}^+
17	D_{21}^+	O_{21}^+	O_{20}^+	O_{20}^+
18	O_{21}^+	D_{22}^-	O_{21}^+	O_{21}^+
19	O_{22}^-	O_{22}^-	O_{22}^-	O_{22}^-

Appendix C.2. Exact rational centres

The following tables specify every coordinate of the four centres exactly. They are rational reference points for uniform estimates; no optimality or exact contact identity is assumed at

a centre. The twelve common coordinates are

$$\begin{aligned}(c_j)_k = K_* := \frac{518617093}{250000000}, \qquad (c_j)_r = R_* := \frac{2066258539}{10^9},\\ (c_j)_w = -R_*,\\ (c_j)_p = \frac{726612331}{500000000}, \qquad (c_j)_\beta = \frac{617532517}{10^9},\\ (c_j)_{v_0} = -\frac{581690673}{10^9}, \qquad (c_j)_{q_0} = -\frac{159528331}{250000000},\\ (c_j)_e = (c_j)_{u_0} = (c_j)_{x_0} = 1, \qquad (c_j)_\sigma = (c_j)_\tau = 0.\end{aligned} \tag{C.7}$$

For a compact exact table of the other twelve coordinates, define

$$\begin{aligned}a_0 &= \frac{10833981}{62500000}, & a_1 &= \frac{47361589}{10^9},\\ a_2 &= \frac{194787661}{250000000}, & a_3 &= \frac{226612331}{500000000},\\ a_4 &= \frac{194712659}{200000000}, & a_5 &= \frac{194705107}{10^9},\\ a_6 &= \frac{639940483}{500000000}, & a_7 &= \frac{175952653}{200000000}.\end{aligned} \tag{C.8}$$

Table C.4: The remaining coordinates of the four rational centres.

Coordinate	c_0	c_1	c_2	c_3
A	$-a_0$	$-K_*$	$-a_0$	$-K_*$
B	$-K_*$	$-a_0$	K_*	a_0
c	1	1	$-a_1$	a_1
d	$-a_1$	a_1	1	1
u_1	a_2	a_2	a_3	$-a_3$
u_2	a_3	$-a_3$	a_2	a_2
x_1	a_4	a_4	1	-1
x_2	1	-1	a_4	a_4
v_1	a_3	a_5	a_3	a_5
v_2	a_5	a_3	$-a_5$	$-a_3$
q_1	$-a_6$	a_7	$-a_6$	a_7
q_2	a_7	$-a_6$	$-a_7$	a_6

In particular, $(c_j)_r + (c_j)_w = 0$, and $R_* - R = 2065458539/10^9$. A full outer box allows every coordinate, including p, to vary independently through its radius R. Only the evolution from an individual source fixes that source's value of p.

Appendix C.3. Finite rational verification of the differential bounds

The companion proof-data table `two_gap_certificate.json` contains the four 23×23 rational matrices C_j, as well as the coordinate order, centres, and selected row labels. Its data schema is `V36_TWO_GAP_FLOW_V1`. Rows of C_j have the order of y in (C.6), and columns have the order of $\Phi_{j,p}$ in (M68). The mathematical role of this finite table is completely specified by the inequalities below; the file name or its hash is not a substitute for those checks.

Each source slack has degree at most two. Thus J_j is affine in ξ, as is $E_j = I - C_j J_j$. For an affine entry $E_{j,ab}$, the exact whole-box bound used is

$$\sup_{\xi \in Q_j} |E_{j,ab}(\xi)| \le |E_{j,ab}(c_j)| + R \sum_{m=0}^{23} |\partial_{\xi_m} E_{j,ab}| \,. \tag{C.9}$$

After summing across each row and taking the maximum, the right side is η in (M72) for each of the four charts. The sum in (C.9) includes the p coordinate. This is the explicit uniform parameter check, rather than a computation at four parameter samples.

Since $\|I - C_j J_j\|_\infty < 1$, the Neumann series makes $C_j J_j$ invertible. Both matrices are square, so J_j is invertible and

$$J_j^{-1} = (C_j J_j)^{-1} C_j.$$

For $g \in \{\sigma, \tau\}$, put $v_{j,g}^0 = C_j e_g$ and $n_{j,g} = \left\| v_{j,g}^0 \right\|_\infty$. The exact inverse direction $v_{j,g} = J_j^{-1} e_g$ obeys

$$\left\| v_{j,g} - v_{j,g}^0 \right\|_\infty \le \frac{\eta n_{j,g}}{1-\eta}, \qquad \|v_{j,g}\|_\infty \le \kappa_{j,g} := \frac{n_{j,g}}{1-\eta}. \tag{C.10}$$

The last two rows of $J_j v_{j,g} = e_g$ give exactly

$$((v_{j,\sigma})_\sigma, (v_{j,\sigma})_\tau) = (1, 0), \qquad ((v_{j,\tau})_\sigma, (v_{j,\tau})_\tau) = (0, 1).$$

Using $F = r - w - \sigma - \tau + \sigma\tau/r$, differentiation gives

$$D_y F \, v_{j,\sigma} = (v_{j,\sigma})_r - (v_{j,\sigma})_w - 1 + \frac{\tau}{r} - \frac{\sigma\tau}{r^2}(v_{j,\sigma})_r. \tag{C.11}$$

For $v_{j,\tau}$, interchange σ and τ in the single-gap term. On the full outer box, $r \ge r_{\min} > 0$ and $|\sigma|, |\tau| \le R$, even when a point of the box is not physical. Consequently the following rational number is a valid upper bound in either case:

$$\begin{aligned} U_{j,g} = (v_{j,g}^0)_r - (v_{j,g}^0)_w - 1 + \frac{2\eta n_{j,g}}{1-\eta} + \frac{R}{r_{\min}} \\ + \frac{R^2 n_{j,g}}{(1-\eta) r_{\min}^2}. \end{aligned} \tag{C.12}$$

For clarity, all eight directional results can be specified by four speed numerators and four height bounds. Set

$$D_v = 9603274520441553881$$

and

$$\begin{aligned} H_0 &= -\frac{28753988041084852657186021690895856248996134979}{10242177916635756600628671715192976500250000000}, \\ H_1 &= -\frac{12333488469648866145304175510636026684107958389}{1024217791663575660062867171519297650025000000}, \\ H_2 &= -\frac{15527152582373713523414265531661512713066376699}{20484355833271513201257343430385953000500000000}, \\ H_3 &= -\frac{42261288828547191891678705825144684910331777797}{12802722395794695750785839643991220625312500000}. \end{aligned}$$

Substitution of the rational data into (C.10) and (C.12) gives Table C.5. Every speed in the table is less than 3, and each H_i is at most $-1/10$.

It remains to verify the unselected physical inequalities on the entire box. If a source polynomial is $P(\xi) = \sum_{|\nu| \le 2} a_\nu \xi^\nu$, its quadratic Taylor expansion gives the rational lower bound

$$P(\xi) \ge P(c_j) - R \sum_{m=0}^{23} |\partial_{\xi_m} P(c_j)| - R^2 \sum_{|\nu|=2} |a_\nu| \qquad (\xi \in Q_j). \tag{C.13}$$

The 23 selected coordinate polynomials are distinct. For every one of the remaining $104 - 23 = 81$ source polynomials, the right side of (C.13) is at least μ in (M74), in each chart. This supplies the physical interior margin used during continuation. The selected twenty band slacks and $r + w$ do not need an interior margin: their nonnegative initial values are conserved exactly, and the last two selected coordinates are the explicitly decreasing nonnegative gaps. All comparisons in this subsection are comparisons of rational numbers.

Table C.5: Exact bounds for the eight inverse-coordinate directions.

Chart	Direction	$D_v\kappa_{j,g}$	$U_{j,g}$
0	σ	14668693972000000000	H_0
0	τ	24566153053000000000	H_1
1	σ	12393729462000000000	H_2
1	τ	10000000000000000000	H_3
2	σ	10000000000000000000	H_3
2	τ	12393729462000000000	H_2
3	σ	24566153053000000000	H_1
3	τ	14668693972000000000	H_0

Appendix C.4. Qualification of a closed source box

For a closed rational enclosure $Q = \prod_{m=0}^{23}[L_m, U_m]$, define

$$d_j(Q) = \max_m \max\{|L_m - (c_j)_m|, |U_m - (c_j)_m|\}.$$

An immediately checkable sufficient guard for this transport is

$$L_r > 0, \qquad L_\sigma, L_\tau \geq 0, \qquad d_j(Q) + 3(U_\sigma + U_\tau) < R. \tag{C.14}$$

Every complete physical source represented by such an enclosure satisfies (M70). The enclosure itself need not consist of physical points; the conclusion is conditional on the complete source inequalities, and does not infer simultaneous feasibility from separate coordinate intervals.

For a proper negative-diagonal source described instead by (r, s, t), the conversion is the actual affine substitution $w = -r, \sigma = r - s, \tau = r - t$. For example, the valid enclosing interval for σ is $[L_r - U_s, U_r - L_s]$, and similarly for τ. This conversion may overestimate the image and may therefore fail the sufficient guard; it does not justify replacing the image by an unrelated smaller box. If (C.14) is not satisfied, including its equality boundary, this transport rule supplies no terminal there. The complete proof graph must retain that region for another qualified rule or for further subdivision. Zero gap faces are included whenever the strict guard holds, because a zero gap requires no evolution in its corresponding direction.

Appendix D. Source accounting and reproducibility

Appendix D.1. Three levels of coverage

The numbers 54, 566 and 469 refer to different levels of the proof. A terminal is a local region with an accepted conclusion. A constituent source may require a tree of such terminals. An original parent responsibility may in turn require many constituent sources. In particular, the last 54-terminal anchor of Section Appendix F discharges one constituent-source obligation, not 54 original parents.

The four residual original-root responsibilities have the following bindings. The indices identify nodes in the retained root; the binary paths identify those same nodes independently of the constituent source numbering.

Box	Original index	Original-root path	Constituents
Q_1	338726	`10011000010101010`	111
Q_2	435003	`10111000001100011000010111011`	240
Q_3	563285	`1101000101111`	159
Q_4	675104	`1110101001111`	56

Accordingly,

$$111 + 240 + 159 + 56 = 566$$

constituent sources discharge four original obligations,

whereas the original-parent accounting is

$$465 \text{ previously discharged obligations} + 4 = 469.$$

There is no proposed bijection between the 566 constituents and the 469 original parents. For each of the four parents, the constituent target-path set must equal its complete frozen source-path set, with no omission or duplicate. Each target also retains its mathematical acceptance and original binding evidence. Counts alone would not establish any of these assertions.

The final anchor within Q_2 contains 48 contradiction terminals, three height or transport terminals, and three canonical-source threshold exclusions. These are the same 54 positions under the terminal semantics of Section M8. Combining them proves safety at α, not that every region is empty at γ. Local paths are interpreted relative to their frozen source and its recorded transformations. In particular, a local path is not silently concatenated with an original-root path to manufacture a source binding.

Appendix D.2. Why the original root still contains four open records

The retained root has 749,693 nodes: 374,846 splits, 374,839 contradiction terminals, four α-safe terminals and four original open records. Its replay proves exactly this partial cover. The four source-bound parent proofs above supply the missing conclusions. The final overlay identifies each parent with the corresponding original-root path and closed rational box, and checks the identities of the accepted parent records. Applying the mixed coverage lemma then gives an effective open count of zero.

The original root is preserved byte for byte. Its four open records are neither deleted nor relabelled as contradiction terminals. This separation permits an independent reader to replay the original tree, replay the additional mathematical arguments and verify their precise attachment points. A root replay that permits the four recorded open positions is therefore an intermediate check, not acceptance of an incomplete final proof.

Appendix D.3. Locating the numerical example and its ancestors

The public release provides the 18.29 MB archive `rho5-final-anchor-example.zip`. The input for (F.2) is `rho5_v53_exact/proof/TERMINAL_DETAILS.json`; the corresponding `SOURCE_CERTIFICATE.json` and `DEPTH19_COMPLETE_SOURCE_PROJECTION.json` bind it to the complete depth-19 high source. The terminal-details file has SHA-256

`11d3e273fff5673cc9cb3ed9673860f388e685bd7ae3b769de0f89d39e20ecc1`.

The archive contains all 49 weighted rows and all 67 coordinate bounds used in Appendix Appendix F, not only its two displayed rows.

The included restoration and joint-verification entry points replay the complete 54-terminal anchor. Their successful result discharges only that anchor; the standalone whole-parent and whole-domain flags remain false. The complete final-cover archive additionally contains the frozen 239-source baseline, the completed parent record and the original-root overlay. The Supplementary Index, S7, identifies these objects and their hashes. Thus the source-to-root chain for the example is

depth-19 source and its contradiction
$\longrightarrow$ complete 54-terminal anchor at `10011101`
$\longrightarrow$ source 240 of 240 in parent 435003
$\longrightarrow$ Q_2 at its recorded original-root position.

The arrows denote complete source-cover compositions, not implications from a terminal count or a receipt name.

Appendix D.4. Independent verification and the supplementary map

The fixed public release is available at `https://github.com/hkjtsgmc79-boop/rho5-proof/releases/tag/v1.0.1`. Its index links all 24 original release assets; the archive inventory is approximately 4.494 GB. The repository's `docs/DOWNLOADS.md` gives sizes and archive identities, `docs/VERIFY.md` gives the existing executable entry points, and `docs/PROOF_MAP.md` distinguishes the written proof, exact acceptors and the companion Lean statements. Large-file transport parts reconstruct the original archive bytes and do not partition the mathematical responsibility.

The supplementary index is organized by mathematical dependency:

Part	Responsibility and verification material
S1	Exact constant, isolated root, attaining parameters and matrix.
S2	Early pivot bounds, complete lift and saturation, including the zero-pivot qualifications.
S3	Canonical representative, reconstruction and canonical height; the V24–V28 upstream dependencies.
S4	All seven complete-X certificate components and their respective frozen entry points.
S5	Six-variable elimination in the negative-diagonal domain and the source model used by the capacity certificates.
S6	Necessary-row, height and transport acceptance rules, with their distinct thresholds and conclusions.
S7	The preserved root, four parent bindings, the final anchor and source-cover composition.
S8	The analytical, upstream, X and negative-diagonal archive collections, replay commands and scope of reproduction.

Dependency order for a complete reproduction.. The following stages follow mathematical dependencies, rather than the chronological labels used during discovery.

1. Fix the polynomial and attaining-matrix inputs (S1), and check the analytic source reductions and their zero and tie cases (S2).
2. Check the canonical representative and reconstruction inputs, including the nested exclusion trees, and then the complete localization and canonical-height certificates (S3).
3. Check all seven complete-X components (S4), using the preceding height theorem for their qualified local exits. Assemble their stated domains into the complete X bound.
4. Check the seventeen-coordinate negative-diagonal source model and the acceptance rules (S5–S6). Replay the preserved original root, retaining its four explicitly identified open positions.
5. Replay the mathematical contributions to each of the four residual parents: their inherited closed complements, verified continuations, and height or transport certificates. Include the final 54-terminal anchor. Match every contribution to its actual parent source and ancestor transformations.
6. Compose the four complete parent covers with the preserved root (S7). Combine the resulting negative-diagonal bound with the complete X bound, saturation and attainment.

The dependency direction in the middle three stages is canonical height, then complete X, then negative-diagonal safety. The final negative-diagonal conclusion is not an input to its own upstream canonical-height proof. The frozen component manifests must be followed recursively; neither the release's asset count nor the final composer's 357 direct file dependencies gives the full transitive mathematical dependency set.

Acceptance conditions.. At each stage, retain the input manifest, verifier identity, exact command or API call, software versions, exit status, mathematical conclusion and elapsed time. A contradiction terminal requires its nonnegative weights and rational upper/lower bounds to be checked. A conditional row requires its guard on the whole box, and a height or transport terminal requires its stated source qualifications and threshold. Such a terminal concludes safety at α; it is not a contradiction at the smaller rational threshold γ.

For the preserved root, the accepted intermediate result has exactly 749,693 nodes, 374,839 contradiction terminals, four α-safe terminals and the four stated open positions. Each completed parent must cover its entire frozen target-path set without omissions or duplicates: respectively 111, 240, 159 and 56 constituent sources. The final composition must match all four parent boxes and paths to the actual root and obtain effective open count zero. Counts are cross-checks on the cover; they do not replace these source and mathematical checks. A missing dependency, failed guard, changed model, incomplete target set or unmatched source invalidates the proposed acceptance at that stage.

Executable scope and implementation trust.. The Supplementary Index, S8, distinguishes command-line entries from lower-level mathematical interfaces and from record composers. The small anchor has a self-contained replay entry. The original negative-diagonal root also has a frozen mathematical replay command. The four-parent chain uses separate mathematical interfaces followed by record composition; its historical configurations include absolute workspace paths. Reproduction on another installation requires an explicit location mapping and fresh output locations while preserving all mathematical input identities and source checks. The dated four-parent orchestration described in Section M11 and Supplementary Index S8 supplies such mappings for the recorded negative-diagonal replay. This supplements the historical lower-level interfaces; it does not regenerate their entire upstream dependency chain. There is not yet a tested location-independent driver that performs this entire transitive chain in one fresh environment. This packaging limitation does not negate the documented component runs and final composition; it limits the presently demonstrated portability of their complete re-execution.

The full computational claim trusts the implementations of the exact rational and interval acceptors, their parsers and row/model generation, and the code that checks source-cover composition. In particular, exact arithmetic prevents rounding uncertainty but does not by itself prove that a guard was applied to the correct source or that all branches were covered. The written arguments in this article establish the intended mathematical rules; the frozen Python/C++ programs implement their finite instances. Independent reimplementation of those rules would reduce reliance on these particular implementations. No discovery optimizer, floating-point search output or search heuristic is accepted as a proof terminal. Python assertion optimization must remain disabled during replay.

The exact Python and C++ acceptors are not asserted to be formally verified programs. Their correctness, the written analytic arguments and the source/domain composition remain part of this proof's trust boundary. The companion first-phase Lean development records its remaining global safety hypotheses explicitly; a successful build or a certificate hash does not remove those hypotheses.

Appendix E. Complete cross-saturated sources and room interfaces

For (M16), the actual core matrix is

$$D = \begin{pmatrix} k & A & B \\ ck & r + cA & r + cB \\ dk & r + dA & w + dB \end{pmatrix}. \tag{E.1}$$

All of (M9)–(M12) and $\|D\|_{\max} \le k$ are retained. No rank, cofactor-sign, or auxiliary entry condition is assumed.

Proposition Appendix E.1 (Certified complete X bound)**.** *Every real complete lift with* (E.1)*,* $p, k, r > 0$*, and* $|w| \le r$*, satisfies* $F = r - w \le \alpha$*.*

The proposition is assembled from four groups of certified source domains. Appendix Appendix A and the Supplementary Index, S4, identify the component certificates and their source domains.

Complete source domain	Certified bound
$0 < k \le 2$	$F < 1653/400$
$k \ge 21/10$	$F < \gamma$
$2 < k \le 21/10,\ r > k$	$F < \gamma$
$\gamma/2 \le k \le 21/10,\ r \le k,\ F \ge \gamma$	$F \le \alpha$

Assembly. The first three rows leave $2 < k < 21/10,\ r \le k$. If $F < \gamma$, (M3) suffices. Otherwise $\gamma \le F \le 2r \le 2k$, hence $k \ge \gamma/2$, and the last row applies. The boundary $r = k$ lies in the last row; $k = 2, 21/10$ have closed coverage. Internal sign and transpose normalizations are actual matrix operations imposing no extra rank or cofactor hypothesis. □

The small-k certificate has 268817 nodes, 134409 contradiction leaves, and no open leaf. The accepted two low-r roots have the following counts:

Root	Nodes	Contradiction leaves	α-safe leaves	Open
I	3450095	1725047	1	0
II	5920895	2873490	86958	0

The counts identify accepted objects, rather than replacing their acceptance rules and domain maps. An α-safe leaf can contain feasible points; it is not an infeasibility certificate at γ.

Order of the analytic inputs.. The strict- and weak-room statements, their full physical predicates, and the qualified 22-coordinate wall entrance are given in Section Appendix E.1. They use the independent canonical-height Theorem M6.9, with detailed arguments in Appendix Appendix B. The 24-coordinate two-gap transport of Lemma M8.1 instead uses the complete X bound; it is not an input to this proposition.

Appendix E.1. The positive-room interfaces and the complete source cover

Here the variables are those of the complete lift (M7), with the core (E.1). Thus $v, q \in \mathbb{R}^3$ are the actual receiver vectors, and $P_j = \beta v_j + q_j$. Write $u = (U, V, T)^T$, $x = (X, Y, Z)^T$, and define

$$\begin{aligned} \delta = U - \beta X, \qquad \Gamma = V - \beta Y, \qquad \nu = \beta Z - T, \\ \tau = UY - XV, \qquad \Delta = UZ - TX, \qquad \mu = VZ - TY. \end{aligned} \tag{E.2}$$

These are room cofactors; Γ is distinct from the height threshold γ, and τ here is distinct from the tail-gap coordinate used in the two-gap transport. A *strict positive-room source*, denoted $\mathcal{R}_0^+$, satisfies all of (M9)–(M12), $\|D\|_{\max} \le k$, and

$$\begin{aligned} p, k, r > 0, \qquad |w| \le r, \qquad A, B, d \le 0, \\ \beta, U, V, T, X, Y, Z > 0, \qquad \delta, \Gamma, \nu, \tau, \Delta, \mu > 0. \end{aligned} \tag{E.3}$$

Here "strict" refers to the seven receiver signs and six cofactor signs; physical pivot ties and the weak bounds on A, B, d are retained. The physical bands retain every upper bound on these coordinates. The *weak positive room* $\mathcal{R}_0^{\ge 0}$ replaces only the seven receiver and six cofactor positivity conditions by nonnegativity. It retains $p, k, r > 0$, $A, B, d \le 0$, and all physical bands. Neither room imposes an additional rank hypothesis.

Proposition Appendix E.2 (Strict-room reduction and weak-room closure)**.** (i) *Every source in $\mathcal{R}_0^+$ with $F \geq 33/8$ has a complete physical representative in $\mathcal{K}$ of Definition Appendix B.1 with exactly the same height. Consequently every source in $\mathcal{R}_0^+$ satisfies $F \leq \alpha$.*

(ii) *Every source in $\mathcal{R}_0^{\geq 0}$ is a limit of complete strict-room sources whose heights converge to its height. Consequently it also satisfies $F \leq \alpha$, including when receiver coordinates, receiver vectors, or cofactors vanish.*

Reduction and closure inputs. For (i), the two-face normalization preserves the complete physical conditions and F, and sends every strict-room source with $F > 4$ to one of

$$e = 1,\ P_1 = -1, \qquad \text{or} \qquad e = 1,\ P_2 = 1,\ v_2 < 0.$$

It changes only e, β, preserving D, p, u, x, v, q and the strict cofactor signs. The second face, including its boundary $v_2 = 0$, has $F < 33/8$. On the first face the feasible shear

$$p' = p + \lambda, \quad \beta' = \beta + \lambda, \quad u' = u + \lambda x, \quad q' = q - \lambda v,$$
$$\lambda = \min\left\{1 - \beta, \frac{1-U}{X}, \frac{1-V}{Y}, \frac{1-T}{Z}\right\}.$$

preserves $D, F, e = 1, P_1 = -1$, every physical band and all six cofactors. To identify the endpoint, $F > 4$ and $|w| \leq r$ give $r > 2$, while $r + dA \leq k$ and $dA \geq 0$ give $k \geq r > 2$. Put $O_{00} = k + Xq_0 + Uv_0$. The physical bands and $\delta > 0$ imply

$$\begin{aligned} 1 + U + (1-\beta)X - k &= (1 - O_{00}) + X(1 + P_0) + \delta(1 + v_0) \\ &\geq 0. \end{aligned}$$

Thus $k \leq 3 - \beta$, and every such high-value source has $\beta < 1$. This also applies to the physical shear endpoint, so $\beta' < 1$. Moreover $\nu' = \beta' Z - T' > 0$, with $\beta', Z \leq 1$, gives $T' < 1$. At least one of the four bounds defining λ is attained; hence the endpoint has $U' = 1$ or $V' = 1$. This includes tied minima and $\lambda = 0$. The latter face also has $F < 33/8$; thus a source with $F \geq 33/8$ reaches $U' = 1$, $0 < V' < 1$. Here p is allowed to change in the shear. The exact dictionary to Definition Appendix B.1 is

$$t = c, \quad E = -d, \quad a = -A, \quad b = -B,$$
$$\widehat{v} = (v_0, v_1, v_2)^T, \quad \widehat{q} = (q_0, -1 - \beta v_1, q_2)^T.$$

The canonical scalars g, h, j are respectively $-v_0, v_1, -q_0$; the scalars denoted v, q there are v_2, q_2 here. This is a single complete matrix at each stage. Theorem M6.9 now gives the bound; sources below $33/8$ already lie below α. The original two-face, tail-face and middle-face statements are V10, Sections 2–6; V11, Theorem T11; and V12, Section 8, respectively, as indexed in Supplementary Index S4.

For (ii), the weak-room approximation theorem (Supplementary Index S4; archived as V27, Theorem 4.1) first applies a positive diagonal contraction that preserves all weak signs and gives strict margins in the physical bands changed by the subsequent regularization. Its height is $\lambda_0^2 F$, with $0 < \lambda_0 < 1$ tending to one. It then regularizes the four ordered nonnegative rays

$$(U, X), \quad (V, Y), \quad (\beta, 1), \quad (T, Z),$$

whose six ordered determinants are precisely (E.2). The regularization makes all receiver coordinates and all six determinants positive, including at zero vectors, while retaining the contracted p, D, e, v, q and hence its height. Reconstructing (M7) preserves the physical bands by their strict margins. Taking the contraction and regularization parameters to their limits gives the stated complete physical approximation and the bound from (i). This closure uses all six weak cofactors. Three adjacent weak minors, or the nonnegative version of a strict ten-term entry test, do not suffice. No same-height strict-room representative is asserted for an arbitrary weak-room source. □

Lemma Appendix E.3 (Qualified complete X-to-room wall entrance)**.** *Use the ordered native coordinates*

$$z = (k, r, w, A, B, c, d, p, e, \beta, u_0, u_1, u_2, x_0, x_1, x_2, v_0, v_1, v_2, q_0, q_1, q_2).$$

Let $\bar{c}_j$, $j = 0, 1, 2, 3$, be the first 22 coordinates of the rational centres in Appendix Appendix C.2, in the declared ξ-order of (M63) rather than the grouped display order of the centre tables. If a complete physical X source satisfies

$$\|z - \bar{c}_j\|_\infty + 3(r + w) < \frac{1}{1250}, \tag{E.4}$$

there is a complete physical path to $r + w = 0$, followed by a signed permutation of the last two rows and columns, whose complete endpoint $\tilde{z}$ belongs to $\mathcal{R}_0^+$ and satisfies

$$F(\tilde{z}) \geq F(z) + \frac{r + w}{10}.$$

Thus $F(z) \leq \alpha - (r + w)/10$.

Wall-flow input. The 22-coordinate V35 certificate and its Round47 flow-tube statement in Supplementary Index S4 keep the source's p and twenty selected physical slacks fixed, decrease $r + w$ at unit speed, and certify speed less than 3 and height gain at least $1/10$. All other physical slacks stay positive on the radius-$1/1250$ outer box. Condition (E.4) keeps the entire path strictly inside this box, so the ordinary differential equation continues to the wall. Only there is the signed tail permutation applied. Its sixteen certified strict conditions are the seven positive receiver coordinates, $-A, -B, -d > 0$, and all six positive cofactors; these imply (E.3). Signed permutations preserve the physical bands and height. Proposition Appendix E.2 then applies. When $r + w = 0$, the path has zero length and the same endpoint argument applies. The use of the centres above merely identifies the common rational data: this proof uses the original 22-coordinate wall certificate. □

Coverage and direction of dependence.. The wall entrance is a qualified local exit. The complete low-r trees also have weak-room exits using Proposition Appendix E.2(ii), direct rational height exits, and exact contradiction leaves. Their original-root cover retains all other signs, zero receivers and cofactors, and physical ties; these are not assumed to enter a positive room. The two complete low-r roots, together with the first three domains of Proposition Appendix E.1, supply the complete cover. In particular $r = k$ remains in the low-r root, and the interfaces $k = 2, 21/10$ are covered as stated in its assembly. Earlier zero pivots have zero subsequent growth by Section M3.

The dependency is the canonical-height theorem, then the room bounds, then the wall exits and complete X cover. The 24-coordinate two-gap transport of Lemma M8.1 instead preserves $r + w$, removes the two cross gaps, and uses Proposition Appendix E.1 to bound its endpoint. The later V43 guard-preserving flow also uses that proposition. Neither is an input to the room bounds or the complete X certificate.

Appendix F. A complete numerical contradiction example

We give an actual terminal from the final negative-diagonal computation. The source being excluded consists of the canonical high sources at this target satisfying $F \geq \gamma$, with the complete parent and ancestor restrictions retained. It is not the entire geometric box. The frozen construction supplies 24 rationally bounded source coordinates and 43 shared products, hence a lifted vector $z \in \mathbb{R}^{67}$, together with 692 necessary linear inequalities. Every physical source under consideration has a lift satisfying these inequalities and all 67 coordinate bounds. The identifiers locate this source in the frozen certificate:

Identifier	Value
Original residual parent	435003
Anchor path	`10011101`
Local terminal path	`1001110110011011111`

The certificate selects 49 rows with positive integer weights; all unselected weights are zero. Two of its actual rows are

Row index	Necessary inequality	Weight
2	$-z_0 - z_4 \le 0$	31 317 836
9	$-z_0 + z_2 + z_{36} \le 0$	229 060 761

Indices here are zero-based. The first 24 lifted coordinates are the source coordinates; in this dictionary $z_{36} = z_5 z_6$. The two displayed rows are an extract, not a two-row certificate: the following calculation uses all 49 selected rows.

Write their exact weighted sum as $a \cdot z \le B$. If $[L_i, U_i]$ is the bound for coordinate i, then

$$m = \sum_{i=0}^{66} \begin{cases} a_i L_i, & a_i \ge 0, \\ a_i U_i, & a_i < 0 \end{cases} \quad \text{satisfies} \quad m \le a \cdot z. \tag{F.1}$$

For example, the actual first coordinate has

$$[L_0, U_0] = \left[\frac{142554217571}{68719476736}, \frac{142566885919}{68719476736}\right],$$
$$a_0 = -\frac{237458354877801951091349245551}{12676506002282294014967032053 76}.$$

Thus its contribution to m uses U_0. Summing the 67 contributions and subtracting the exact weighted right-hand side gives

$$m - B = \frac{1005495384602754603811064507351680469029685784938386479}{5444517870735015415413993718908291383296000000000000} > 0. \tag{F.2}$$

Consequently $m \le a \cdot z \le B < m$ is impossible. This is an exact rational contradiction for the specified high source. Its margin is approximately 1.84680335×10^3 in the displayed weight normalization. It depends on that normalization and is not a bound on $\alpha - F$.

The complete row list, weights, coordinate bounds and weighted sums are supplied in the real example identified in Appendix Appendix D. Its source record also identifies the frozen parent, projection, ancestor trace and target image. These bindings are needed to establish that the rows and bounds apply to this source; the arithmetic in (F.2) alone does not establish that fact.

This terminal discharges one position in the anchor's complete 54-terminal cover. The other positions are retained, including those proved safe by a height bound or feasible transport instead of a contradiction. By Lemma M8.2, their composition proves that complete anchor safe at α. The anchor then discharges the last of 240 constituent sources of parent 435003; the other 239 sources retain their accepted coverage. Finally, that parent is bound to one of the four open positions of the preserved original root. Thus the example illustrates both parts of certificate acceptance: an exact local inequality and a complete, source-preserving route back to the original domain.

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